The `revtex4-1` document class of the American Physical Society *

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Version 4.1s, dated 2020/09/30

This file embodies the implementation of the APS REVTEX 4.1 document class for electronic submissions to journals.

The distribution point for this work is http://publish.aps.org/revtex4/, which contains fully unpacked, prebuilt runtime files and documentation.

Contents

1 Using REVTEX **2**
1.1 Bill of Materials . . . 2
1.1.1 Primary Source . . . 2
1.1.2 Generated by `latex ltxutil.dtx` . . . 3
1.1.3 Generated by `tex ltxutil.dtx` . . . 3
1.1.4 Auxiliary . . . 3

2 Code common to all modules **3**

3 The driver module `doc` **4**
3.1 The Preamble . . . 4
3.1.1 Docstrip and info directives . . . 4
3.2 The "Read Me" File . . . 5
3.3 The Document Body . . . 8

4 Using this package **8**
4.1 Invoking the package . . . 8

5 Compatibility with LATEX's Required Packages **9**
5.1 array . . . 9
5.2 longtable . . . 9

*Work under hire to The American Physical Society.

†First revision of REVTEX4.0 (unreleased) by David Carlisle

6 Implementation of package **9**
6.1 Beginning of the `package` DOCSTRIP module . . . 9
6.2 Banner and beginning of the `kernel` DOCSTRIP module . . . 10
6.3 Errors and warnings . . . 10
6.4 New Tools . . . 11
6.5 Boolean Control . . . 12
6.6 Begin Document Structure . . . 14
6.7 Class Extensions . . . 18
6.8 Type Tools . . . 20
6.9 Display Math . . . 21
6.10 Footnotes . . . 23
6.11 Floats . . . 28
6.11.1 Usage notes . . . 28
6.11.2 Robustifying fragile commands . . . 29
6.11.3 Preparing for the `hyperref package` . . . 30
6.11.4 Footnotes within floats, unfloating floats, float font . . . 30
6.11.5 Writing floats out to a file . . . 33
6.12 Counters . . . 37
6.13 Customization of Sections . . . 37
6.14 Patch the `tabular` and `array` Environments . . . 41
6.15 Repair other broken parts of LaTeX . . . 63
6.16 Syntax . . . 63
6.17 Auto-indented Contents . . . 63
6.18 Lists . . . 67
6.19 Hypertext capabilities . . . 68
6.20 End of the `kernel` DOCSTRIP module . . . 71

Index **72**

1 Using REVTEX

The file `README` has retrieval and installation information.

User documentation is presented separately in `auguide.tex`.

The file `template.aps` is a boilerplate file.

1.1 Bill of Materials

Following is a list of the files in this distribution arranged according to provenance.

1.1.1 Primary Source

One single file generates all.

```
%ltxutil.dtx
%
```

1.1.2 Generated by `latex ltxutil.dtx`

Typesetting the source file under pdflatex generates the readme and the documentation.

```
%00readme ltxutil.pdf
%
```

1.1.3 Generated by `tex ltxutil.dtx`

Typesetting this file with TEX generates the package file.

```
%ltxutil.sty
%
```

1.1.4 Auxiliary

The following are auxiliary files generated in the course of running LATEX:

```
%ltxutil.aux ltxutil.idx ltxutil.ind ltxutil.log ltxutil.toc
%
```

2 Code common to all modules

We want to require only one place in this file where the version number is stated, and we also want to ensure that the version number is embedded into every generated file.

Now we declare that these files can only be used with LATEX 2_ε. An appropriate message is displayed if a different TEX format is used.

```
%<*doc|package>
\NeedsTeXFormat{LaTeX2e}[1995/12/01]%
%</doc|package>
```

As desired, the following modules all take common version information:

```
%<kernel&!package&!doc>\typeout{%
%<*package|doc>
\ProvidesFile{%
%</package|doc>
%<*kernel|package|doc>
ltxutil%
%</kernel|package|doc>
%<*doc>
 .dtx%
%</doc>
%<package>.sty%
%<*package|doc>
}%
%</package|doc>
```

The following line contains, for once and for all, the version and date information. By various means, this information is reproduced consistently in all generated files and in the typeset documentation. Give credit where due.

```
%<*doc|package|kernel>
%<version>
 [2020/09/30 4.1s utilities package (portions licensed from W. E. Baxter web at superscript.com
%</doc|package|kernel>
%<kernel&!package&!doc>}%
```

3 The driver module `doc`

This module, consisting of the present section, typesets the programmer's documentation, generating the `00readme` as required.

Because the only uncommented-out lines of code at the beginning of this file constitute the `doc` module itself, we can simply typeset the `.dtx` file directly, and there is thus rarely any need to generate the "doc" DOCSTRIP module. Module delimiters are nonetheless required so that this code does not find its way into the other modules.

The `\end{document}` command concludes the typesetting run.

```
%<*doc>
```

3.1 The Preamble

The programmers documentation is formatted with the `ltxdoc` class with local customizations, and with the usual code line indexing.

```
\documentclass{ltxdoc}
\RequirePackage{ltxdocext}%
\let\url\undefined
\RequirePackage[colorlinks=true,linkcolor=blue]{hyperref}%
\pdfstringdefDisableCommands{%
  \let\file\relax
  \let\sc\relax
}
\makeatletter
\@ifundefined{package@font}{}%
  {\expandafter\RequirePackage\expandafter{\csname package@font\endcsname}}
\makeatother
\CodelineIndex\EnableCrossrefs % makeindex -s gind.ist ltxutil
\RecordChanges % makeindex -s gglo.ist -o ltxutil.gls ltxutil.glo
```

3.1.1 Docstrip and info directives

We use so many DOCSTRIP modules that we set the `StandardModuleDepth` counter to 1.

```
\setcounter{StandardModuleDepth}{1}
```

The following command retrieves the date and version information from this file.

```
\expandafter\GetFileInfo\expandafter{\jobname.dtx}%
```

3.2 The "Read Me" File

As promised above, here is the contents of the "Read Me" file. That file serves a double purpose, since it also constitutes the beginining of the programmer's documentation. What better thing, after all, to have appear at the beginning of the typeset documentation?

A good discussion of how to write a ReadMe file can be found in Engst, Tonya, "Writing a ReadMe File? Read This" *MacTech* October 1998, p. 58.

Note the appearance of the `\StopEventually` command, which marks the dividing line between the user documentation and the programmer documentation.

The usual user will not be asked to do a full build, not to speak of the bootstrap. Instructions for carrying out these procedures begin the programmer's manual.

```
\begin{filecontents*}{00readme}
\title{%
 A \LaTeX\ Package of utility macros%
 \thanks{%
  This file has version number \fileversion,
  last revised \filedate.%
 }%
 \thanks{%
  Version \fileversion\ \copyright\ 2009--2020 The American Physical Society
 }%
}%
\author{%
 Arthur Ogawa%
 \thanks{\texttt{mailto:arthur\_ogawa at sbcglobal.net}}%
}%
%\iffalse
% For version number and date,
% search on "\fileversion" in the .dtx file,
% or see the end of the 00readme file.
%\fi
\maketitle

This file embodies the \classname{ltxutil} package,
the implementation and its user documentation.

The distribution point for this work is
\url{publish.aps.org/revtex},
which contains prebuilt runtime files, documentation, and full source,
ready to add to a TDS-compliant \TeX\ installation.

The \classname{ltxutil} package was commissioned by the American Physical Society
and is distributed under the terms of the \LaTeX\ Project Public License,
the same license under which all the portions of \LaTeX\ itself are distributed.
```

```
Please see \url{http://ctan.tug.org/macros/latex/base/lppl.txt} for details.

To use this document class, you must have a working
\TeX\ installation equipped with \LaTeXe\
and possibly pdftex and Adobe Acrobat Reader or equivalent.

To install, retrieve the distribution,
unpack it into a directory on the target computer,
and move the file \file{ltxutil.sty}
into a location in your filesystem where it will be found by \LaTeX.

To use, read the user documentation \file{ltxutil.pdf}.

\tableofcontents

\section{Processing Instructions}

The package file \file{ltxutil.sty}
is generated from this file, \file{ltxutil.dtx},
using the {\sc docstrip} facility of \LaTeX
via |tex ltxutil.dtx| (Note: do \emph{not} use \LaTeX\ for this task).
The typeset documentation that you are now reading is generated from
the same file by typesetting it with \LaTeX\ or pdftex
via |latex ltxutil.dtx| or |pdflatex ltxutil.dtx|.

\subsection{Build Instructions}

You may bootstrap this suite of files solely from \file{ltxutil.dtx}.
Prepare by installing \LaTeXe\ (and either tex or pdftex) on your computer,
then carry out the following steps:
\begin{enumerate}
\item
Within an otherwise empty directory,
typeset \file{ltxutil.dtx} with \LaTeX\ or pdflatex;
you will obtain the typeset documentation you are now reading,
along with the file \file{00readme}.

Note: you will have to run \LaTeX, then
\file{makeindex} \texttt{-s gind.ist ltxutil.idx}, then
\file{makeindex} \texttt{-s gglo.ist -o ltxutil.gls ltxutil.glo}, then
\LaTeX\ again in order to obtain a valid index and table of contents.
\item
Now typeset \file{ltxutil.dtx} with \TeX (not \LaTeX),
thereby generating the package file \file{ltxutil.sty}.
\item
Install the following files into indicated locations within your
TDS-compliant \texttt{texmf} tree (you may need root access):
\begin{itemize}
\item
\file{$TEXMF/}\file{tex/}\file{latex/}\file{revtex/}\classname{ltxutil.sty}
```

```
\item
\file{$TEXMF/}\file{source/}\file{latex/}\file{revtex/}\classname{ltxutil.dtx}
\item
\file{$TEXMF/}\file{doc/}\file{latex/}\file{revtex/}\classname{ltxutil.pdf}
\end{itemize}
where \file{$TEXMF/} stands for \file{texmf-local/}, or some other \texttt{texmf} tree
in your installation.
\item
Run \texttt{mktexlsr} on \file{$TEXMF/} (you may need root access).
\item
Build and installation are now complete;
now put a \cmd\usepackage\texttt{\{ltxutil\}} in your document preamble!
\end{enumerate}

\subsection{Change Log}
\changes{4.0b}{1999/06/20}{AO: Fixed spurious \texttt{CR} and (return) characters in output fil
\changes{4.0b}{1999/06/20}{AO: Removed superfluous \cs{def}s, changed to using \cs{floats@sw} a
\changes{4.0b}{1999/06/20}{only execute if there really were floats of the given type}
\changes{4.0b}{1999/06/20}{Support the hack with \cs{prepdef}, and delay until \cs{AtBeginDocum
\changes{4.0c}{1999/11/13}{(AO, 110) Install hooks for endfloats processing}
\changes{4.0c}{1999/11/13}{(AO, 116) Hyperref compatibility}
\changes{4.0c}{1999/11/13}{(AO, 130) Interference from array package}
\changes{4.0c}{1999/11/13}{*-form mandates pagebreak at each float; only print section head if
\changes{4.0d}{2000/04/10}{(AO, 127) Floats placed [h] to allow page breaks}
\changes{4.0d}{2000/04/10}{(AO, 174) kernel fix}
\changes{4.0d}{2000/05/19}{(AO, 224) Hyperref compatibility.}
\changes{4.0d}{2000/05/23}{Allow things to break over pages by setting array@default.}
\changes{4.0e}{2000/11/16}{(AO, 221) Remove samepage command from @xfloat@prep: If the float ca
\changes{4.0f}{2001/07/13}{(AO, 404) Hyperref compatibility}
\changes{4.1a}{2008/01/19}{(AO, 459) do not assume \cs{class@name} is defined}%
\changes{4.1a}{2008/01/19}{(AO, 461) Change the csname from \cs{@dotsep} to \cs{ltxu@dotsep}. T
\changes{4.1a}{2008/01/19}{(AO, 475) I had not properly reproduced the LaTeX macro \cs{eqnarray
\changes{4.1a}{2008/01/19}{(AO, 479) Per: Dylan Thurston<dpt at math.harvard.edu>}%
\changes{4.1a}{2008/06/30}{(AO) Make \cs{addtocontents} a \cs{long} \cs{def}; gobble up \cs{foo
\changes{4.1a}{2008/06/30}{(AO) Remove code that avoided changes to \cs{@xfootnotemark}}%
\changes{4.1a}{2008/06/30}{(AO, 438) Complete rewrite of footnote macros.}
\changes{4.1a}{2008/07/07}{\cs{@xfloat@prep} calls \cs{ltx@footnote@pop} to restore the origina
\changes{4.1a}{2008/08/12}{\cs{class@documenthook} is the last \cs{AtBeginDocument} token now}
\changes{4.1a}{2008/08/12}{Class extension mechanism \cs{@pushfilename@ltx} and \cs{@p@pfilenam
\changes{4.1a}{2008/08/12}{Class extension mechanism \cs{class@extension}, \cs{class@extensionf
\changes{4.1a}{2008/08/12}{Get rid of \cs{set@typesize@hook} \cs{set@pica@hook} and the \cs{nor
\changes{4.1b}{2008/08/12}{(AO, 487) Support for video figures and the \cs{setfloatlink} comman
\changes{4.1b}{2008/08/12}{(AO, 505) try to accommodate \classname{colortbl}.}
\changes{4.1b}{2008/08/12}{Acquire \classname{hyperref} savoire}
\changes{4.1b}{2008/08/12}{Default assignment of \cs{float@sw} now, not at \cs{AtBeginDocument}
\changes{4.1b}{2008/08/12}{If class option \classoption{lengthcheck} is in effect, log the heig
\changes{4.1b}{2008/08/12}{No need to protect against undefined \cs{float@sw}}
\changes{4.1b}{2008/08/12}{Patch the array package even later: after all package patches go in.
\changes{4.1b}{2008/08/12}{Refine toc processing: provide default.}%
\changes{4.1b}{2008/08/12}{Tally and log the height of a float class}
```

```
\changes{4.1d}{2009/03/27}{(AO, 511) Compatability with lineno.sty's erroneous way of detecting
\changes{4.1f}{2009/07/07}{(AO, 515) Hook for setting the font of a footnote}
\changes{4.1f}{2009/07/10}{(AO, 518) Tally register overflow when locument is long}
\changes{4.1g}{2009/10/06}{(AO, 532) Both arguments of \cs{href} get sanitized}%
\changes{4.1g}{2009/10/07}{(AO, 525) Remove phantom paragraph above display math that is given
\changes{4.1g}{2009/10/07}{(AO, 539) Use of double-backslash in argument of \cs{section} gives
\changes{4.1n}{2009/12/05}{(AO, 569) Use of \classname{hyperref} interferes with column balanci
\changes{4.1n}{2009/12/06}{(AO) Incorporate change to ltmiscen.dtx v1.1i 2000/05/19}%
\changes{4.1n}{2009/12/09}{(AO, 569) execute \classname{atveryend}'s \cs{Call@AfterLastShipout}
\changes{4.1n}{2009/12/13}{(AO, 574) protect against \classname{lineno.sty}, which forces a vis
\changes{4.1n}{2010/01/02}{(AO, 571) Interface \cs{set@footnotewidth} for determining the set w
\changes{4.1n}{2010/01/02}{(AO, 571) allow split after last line of footnote}%
\changes{4.1n}{2010/01/06}{(AO, 572) title block footnotes numbered independently from body foo
\changes{4.1p}{2010/02/24}{(AO, 582) A patch of \classname{hyperref.sty} to provide backward co
\changes{4.1s}{2020/09/19}{(PHO) Adapt \cs{document} and \cs{enddocument} hooks to the 2020-10-

\end{filecontents*}
```

3.3 The Document Body

Here is the document body, containing only a `\DocInput` directive—referring to this very file. This very cute self-reference is a common `ltxdoc` idiom.

```
\begin{document}%
\expandafter\DocInput\expandafter{\jobname.dtx}%
\end{document}

%</doc>
```

4 Using this package

Once this package is installed on your filesystem, you can employ it in adding functionality to LaTeX by invoking it in your document or document class.

4.1 Invoking the package

In your document, you can simply call it up in your preamble:

```
%\documentclass{book}%
%\usepackage{ltxutil}%
%\begin{document}
%⟨your document here⟩
%\end{document}
```

However, the preferred way is to invoke this package from within your customized document class:

```
%\NeedsTeXFormat{LaTeX2e}[1995/12/01]%
%\ProvidesClass{myclass}%
%\RequirePackage{ltxutil}%
```

```
%\LoadClass{book}%
%⟨class customization commands⟩
%\endinput
```

Once loaded, the package gives you acccess to certain procedures, usually to be invoked by a LaTeX command or environment, but not at the document level.

5 Compatibility with LaTeX's Required Packages

Certain packages, usually ones written by members of the LaTeX Project itself, have been designated "required" and are distributed as part of standard LaTeX. These packages have been placed in a priviledged position vis á vis the LaTeX kernel in that they override the definitions of certain kernel macros.

The `ltxutil` package will be incompatible with any package that redefines any of the kernel macros that `ltxutil` patches—if that package is loaded *after* `ltxutil`. This means that for greatest compatibility, `ltxutil` should be loaded *after*, say, `ftnright`, which overwrites LaTeX's kernel procedures `\@outputdblcol`, `\@startcolumn`, and `\@makecol`.

Hereinafter follows some notes on specific LaTeX packages.

5.1 array

This package alters the way tabular environments are done, therefore it could run afoul of the LaTeX "required" package `array` or any package that calls for it to be loaded. However, this package has provisions for remaining compatible with `array`. So long as the version of `array` that is used with this package has the appropriate meanings for the procedures it overwrites, all should be well.

5.2 longtable

David Carlisle's `longtable` package modifies both the LaTeX kernel and the `array` package. This package must therefore alter `\LT@array`. For now, that job is handled by `ltxgrid`.

6 Implementation of package

Special acknowledgment: this package uses concepts pioneered and first realized by William Baxter (mailto:web at superscript.com) in his SuperScript line of commercial typesetting tools, and which are used here with his permission.

6.1 Beginning of the package DOCSTRIP module

```
%<*package>
\def\package@name{ltxutil}%
\expandafter\PackageInfo\expandafter{\package@name}{%
 Utility macros for \protect\LaTeXe,
```

```
 by A. Ogawa (arthur_ogawa at sbcglobal.net)%
}%
%</package>
```

6.2 Banner and beginning of the `kernel` DOCSTRIP module

```
%<*kernel>
```

6.3 Errors and warnings

`\class@err` `\class@warn` `\class@info`

A few shorthands for Class messages. Your document class should define `\class@name`.

```
\def\class@err#1{\ClassError{\class@name}{#1}\@eha}%
\def\class@warn#1{\ClassWarningNoLine{\class@name}{#1}}%
\def\class@info#1{\ClassInfo{\class@name}{#1}}%
\def\obsolete@command#1{%
 \class@warn@end{Command \string#1\space is obsolete.^^JPlease remove from your document}%
 \global\let#1\@empty
 #1%
}%
\def\replace@command#1#2{%
 \class@warn@end{Command \string#1\space is obsolete;^^JUse \string#2\space instead}%
 \global\let#1#2%
 #1%
}%
\def\replace@environment#1#2{%
 \class@warn@end{Environment #1 is obsolete;^^JUse #2 instead}%
 \glet@environment{#1}{#2}%
 \@nameuse{#1}%
}%
\def\incompatible@package#1{%
 \@ifpackageloaded{#1}{%
  \def\@tempa{I cannot continue. You must remove the \string\usepackage\ statement that caused
  \ClassError{\class@name}{The #1 package cannot be used with \class@name}%
  \@tempa\stop
 }{%
  \class@info{#1 was not loaded (OK!)}%
 }%
}%
\def\class@warn@end#1{%
 \gappdef\class@enddocumenthook{\class@warn{#1}}%
}%
```

Give `\class@name` a meaning if it does not already have one.

```
\ifx\undefined\class@name
 \def\class@name{ltxutil}%
 \class@warn{You should define the class name before reading in this package. Using default}%
\fi
```

6.4 New Tools

\t@

```
\def\t@{to}%
```

\dimen@iii

```
\dimendef\dimen@iii\thr@@
```

\halignt@

```
\def\halignt@{\halign\t@}%
```

\f@ur Analogous to \@ne, \tw@, and \thr@@.

```
\chardef\f@ur=4\relax
\chardef\cat@letter=11\relax
\chardef\other=12\relax
```

\let@environment
\glet@environment The directive \let@environment takes care of a common programming idiom whereby one environment is made a synonym for another.

```
\def\let@environment#1#2{%
 \expandafter\let
 \csname#1\expandafter\endcsname\csname#2\endcsname
 \expandafter\let
 \csname end#1\expandafter\endcsname\csname end#2\endcsname
}%
\def\glet@environment#1#2{%
 \global\expandafter\let
 \csname#1\expandafter\endcsname\csname#2\endcsname
 \global\expandafter\let
 \csname end#1\expandafter\endcsname\csname end#2\endcsname
}%
```

\tracingplain The command \tracingplain causes TEX's tracing parameters to return to the values set by default. This command is sometimes useful when you have said \tracingall somewhere and want to restore. The \traceoutput command causes \tracingoutput diagnostics upon \shipout.

```
\newcommand\tracingplain{%
 \tracingonline\z@\tracingcommands\z@\tracingstats\z@
 \tracingpages\z@\tracingoutput\z@\tracinglostchars\@ne
 \tracingmacros\z@\tracingparagraphs\z@\tracingrestores\z@
 \showboxbreadth5\showboxdepth3\relax %\errorstopmode
 }%
\newcommand\traceoutput{%
 \appdef\@resetactivechars{\showoutput}%
}%
```

\say
\saythe The commands \say and \saythe cause diagnostic messages in the TEX log that give the value of a control sequence name or a register respectively.

```
\newcommand\say[1]{\typeout{<\noexpand#1=\meaning#1>}}%
\newcommand\saythe[1]{\typeout{<\noexpand#1=\the#1>}}%
```

\fullinterlineskip Resets the \prevdepth so that the full amount of \baselineskip glue will be inserted by the \baselinesklip mechanism. Can be invoked just after a \hrule to undo its default suppression of base line skip.

```
\def\fullinterlineskip{\prevdepth\z@}%
```

\count@i
\count@ii

```
\countdef\count@i\@ne
\countdef\count@ii\tw@
```

6.5 Boolean Control

We introduce just enough of the Boolean calculus for TEX. Alan Jeffrey was the pioneer here, with an article in TUGboat (Vol. 11, No. 2, page 237). This implementation owes a debt to William Baxter (web at superscript.com). See articles by Baxter and Ogawa in the proceedings of the 1994 TUG meeting, TUGboat Vol. 15, No. 3.

\prepdef
\appdef
\gappdef

Provide the capability of performing head- and tail patches. The procedure \prepdef prepends to the given macro the tokens specified in its second argument. Likewise for \appdef, except that it appends. Note that the first 10 toks registers are utility registers, and we simply make a control sequence name, \toks@ii, for one of them.

```
\long\def\prepdef#1#2{%
 \@ifxundefined#1{\toks@{}}{\toks@\expandafter{#1}}%
 \toks@ii{#2}%
 \edef#1{\the\toks@ii\the\toks@}%
}%
\long\def\appdef#1#2{%
 \@ifxundefined#1{\toks@{}}{\toks@\expandafter{#1}}%
 \toks@ii{#2}%
 \edef#1{\the\toks@\the\toks@ii}%
}%
\long\def\gappdef#1#2{%
 \@ifxundefined#1{\toks@{}}{\toks@\expandafter{#1}}%
 \toks@ii{#2}%
 \global\edef#1{\the\toks@\the\toks@ii}%
}%
\long\def\appdef@val#1#2{%
 \appdef#1{{#2}}%
}%
\long\def\appdef@e#1#2{%
 \expandafter\appdef
 \expandafter#1%
 \expandafter{#2}%
}%
\long\def\appdef@eval#1#2{%
 \expandafter\appdef@val
 \expandafter#1%
```

```
  \expandafter{#2}%
}%
\toksdef\toks@ii=\tw@
```

\@ifxundefined
\@ifnotrelax
\@argswap
\@argswap@val

Certain utility procedures use `\@ifxundefined`, which is defined here in terms of `\@ifx`. Others use `\@ifnotrelax`, namely when the control sequence name is manufactured by the use of `\csname`.

The procedures `\@argswap`and `\@argswap@val`are used to facilitate control of expansion.

```
\long\def\@ifxundefined#1{\@ifx{\undefined#1}}%
\long\def\@ifnotrelax#1#2#3{\@ifx{\relax#1}{#3}{#2}}%
\long\def\@argswap#1#2{#2#1}%
\long\def\@argswap@val#1#2{#2{#1}}%
\def\@ifxundefined@cs#1{\expandafter\@ifx\expandafter{\csname#1\endcsname\relax}}%
```

\rvtx@ifformat@geq

Some changes in the LaTeX kernel requires us to conditionally define some macros depending on the version of the kernel. `\rvtx@ifformat@geq` will check if the release date of the currently-running LaTeX 2_ε kernel is greater or equal to the argument (the argument should be in the format `yyyy-mm-dd`).

```
\ifx\IfFormatAtLeastTF\undefined
  \def\rvtx@ifformat@geq{\@ifl@t@r\fmtversion}%
\else
  \let\rvtx@ifformat@geq\IfFormatAtLeastTF
\fi
```

\@boolean
\@boole@def

In order to define `\@ifx`, we first must create the "defining word" (term taken form our Forth vocabulary) `\@boole@def`, which employs `\@boolean` to do its job.

```
\def\@boolean#1#2{%
  \long\def#1{%
    #2% \if<something>
      \expandafter\true@sw
    \else
      \expandafter\false@sw
    \fi
  }%
}%
\def\@boole@def#1#{\@boolean{#1}}% Implicit #2
```

\@booleantrue
\@booleanfalse

The procedures `\@booleantrue` and `\@booleanfalse` are assignment operators for Boolean flags.

```
\def\@booleantrue#1{\let#1\true@sw}%
\def\@booleanfalse#1{\let#1\false@sw}%
```

\@ifx
\@ifx@empty
\@if@empty
\@ifcat
\@ifdim
\@ifeof
\@ifhbox
\@ifhmode
\@ifinner
\@ifmmode
\@ifnum
\@ifodd
\@ifvbox
\@ifvmode
\@ifvoid

We can now invoke the defining word to create the procedures `\@ifx` and friends.

Compatibility Note: earlier versions of this package defined a procedure `\@ifempty`. However, for compatibility with AMSLaTeX, we must avoid the following three names: `\@ifempty`, `\@xifempty`, and `\@ifnotempty`.

```
\@boole@def\@ifx#1{\ifx#1}%
```

```
\@boole@def\@ifx@empty#1{\ifx\@empty#1}%
\@boole@def\@if@empty#1{\if!#1!}%
%\@boole@def\@if@sw#1{\csname if#1\endcsname}%
\def\@if@sw#1#2{#1\expandafter\true@sw\else\expandafter\false@sw#2}%
\@boole@def\@ifdim#1{\ifdim#1}%
\@boole@def\@ifeof#1{\ifeof#1}%
\@boole@def\@ifhbox#1{\ifhbox#1}%
\@boole@def\@ifhmode{\ifhmode}%
\@boole@def\@ifinner{\ifinner}%
\@boole@def\@ifmmode{\ifmmode}%
\@boole@def\@ifnum#1{\ifnum#1}%
\@boole@def\@ifodd#1{\ifodd#1}%
\@boole@def\@ifvbox#1{\ifvbox#1}%
\@boole@def\@ifvmode{\ifvmode}%
\@boole@def\@ifvoid#1{\ifvoid#1}%
```

`\true@sw` `\false@sw` Note that when a Boolean operator expands, it employs two macros that act as selectors, defined here.

```
\long\def\true@sw#1#2{#1}%
\long\def\false@sw#1#2{#2}%
```

`\loopuntil` `\loopwhile` Loop control using the Boolean idiom. Superior to `\loop`...`\repeat` because these can be nested. The tail of the argument must have a Boolean predicate.

```
\long\def\loopuntil#1{#1{}{\loopuntil{#1}}}%
\long\def\loopwhile#1{#1{\loopwhile{#1}}{}}%
```

`\@provide` A defining word that refuses to clobber a prior meaning.

```
\def\@provide#1{%
 \@ifx{\undefined#1}{\true@sw}{\@ifx{\relax#1}{\true@sw}{\false@sw}}%
 {\def#1}{\def\j@nk}%
}%
```

6.6 Begin Document Structure

The standard LaTeX mechanism `\AtBeginDocument` is inadequate because the `\vsize` is bound much too early. We supply here a mechanism whereby decisions about the page layout can be deferred until `\AtBeginDocument` time.

The problem we are working around is that the `\AtBeginDocument` hook in `\document` appears long after the calculation of `\vsize` and `\hsize`, that is, LaTeX provides no mechanism for deferring the decision about the page grid until `\AtBeginDocument` time. We fix things by prepending a hook at the very beginning of `\document`.

As it turns out, though, it appears feasible to simply invoke the desired column grid command at `\AtBeginDocument` time, since the MVL has nothing in it at that time that would be problematical.

`\document` We begin by installing hooks into `\document` that we will manage ourselves.

The 2020-10-01 LaTeX release got a new hook management system and several new hooks (several previously provided by etoolbox). The one we want here is `begindocument/before`, the first thing executed by `\document`, right after ending the group started by `\begin`.

Thus, if the LaTeX kernel date is 2020-10-01 we just add to that hook, otherwise resort to the old method, patching `\document`: end the group started by `\begin`, apply our hook, and conclude our shenanigans by absorbing the first token of the expansion of `\document`, which we assume to be `\endgroup` (true until the aforementioned release).

```
\rvtx@ifformat@geq{2020-10-01}%
  {%
    \AddToHook{begindocument/before}{\document@inithook}%
  }{%
    \prepdef\document{%
     \endgroup
     \document@inithook
     \true@sw{}%
    }%
  }
```

`\document@inithook` To use, simply `\appdef\document@inithook{`⟨*your tokens here*⟩`}`.

```
\let\document@inithook\@empty
```

`\class@documenthook` We install the last `\AtBeginDocument` hook, namely the procedure `\class@documenthook`.
`\class@enddocumenthook` Within the document class, we will use this hook exclusively, so as to avoid interference from other packages. Similarly with `\class@enddocumenthook`, installed via `\AtEndDocument`.

A document class using this package should do as this package does and just say, `\appdef \class@documenthook` instead of `\AtBeginDocument`, and `\appdef \class@enddocumenthook` instead of `\AtEndDocument`.

```
\appdef\document@inithook{%
 \AtBeginDocument{\class@documenthook}%
}%
\AtEndDocument{%
 \class@enddocumenthook
}%
\let\class@documenthook\@empty
\let\class@enddocumenthook\@empty
```

`\enddocument` The standard LaTeX `\end{document}` processing is a potential problem, particu-
`\check@aux` larly when the output routine has been changed by `ltxgrid`. We separate out the
`\do@check@aux` procedure that checks the auxiliary file at the end of the job so that later it can be called from the safety of the output routine. We will do this to ensure that the `\@mainaux` stream is not closed until the last page of the job is shipped out, and that can only be done by coordinating with the output routine.

This approach, however, will only be done for older versions of the LaTeX kernel:

```
\rvtx@ifformat@geq{2020-10-01}{%
```

```
  % <definitions for newer LaTeX later>
}{%
  % <definitions for older LaTeX>
\def\enddocument{%
```

The following line from ltxutil.dtxltmiscen.dtx 'resets `\AtEndDocument`for latex/3060'.

```
 \let\AtEndDocument\@firstofone
 \@enddocumenthook
 \@checkend{document}%
```

The `\clear@document` statement ends the current page (we must guarantee no further shipouts), then executes all cleanup procedures that must occur only after the last shipout. Clients will queue up their procedures via `\AfterLastShipout`, if it exists, otherwise by doing `\appdef\clear@document`.

```
 \clear@document
```

We are very close to ending the TEX run, now.

```
 \check@aux
 \deadcycles\z@
 \@@end
}%
\def\check@aux{\do@check@aux}%
\def\do@check@aux{%
 \@if@sw\if@filesw\fi{%
  \immediate\closeout\@mainaux
  \let\@setckpt\@gobbletwo
  \let\@newl@bel\@testdef
  \@tempswafalse
  \makeatletter
  \input\jobname.aux\relax
 }{}%
 \@dofilelist
 \@ifdim{\font@submax >\fontsubfuzz\relax}{%
  \@font@warning{%
   Size substitutions with differences\MessageBreak
   up to \font@submax\space have occured.\@gobbletwo
  }%
 }{}%
 \@defaultsubs
 \@refundefined
 \@if@sw\if@filesw\fi{%
  \@ifx{\@multiplelabels\relax}{%
   \@if@sw\if@tempswa\fi{%
    \@latex@warning@no@line{%
     Label(s) may have changed.
     Rerun to get cross-references right%
    }%
   }{}%
  }{%
```

```
    \@multiplelabels
  }%
 }{}%
}%
}
```

`\rvtx@enddocument@patch` For newer LaTeX we'll try to be a bit more future-proof (no miracle though). The code for `\enddocument` (in pre-2020-10-01 LaTeX) is roughly:

```
%   \def\enddocument{%
%     ⟨hooks and bookkeeping⟩
%     \clearpage
%     ⟨read main .aux and final checks⟩
%     \@@end
%   }
%
```

and the patches above replace the `\clearpage` by its own `\clear@document`, and `<read main .aux and final checks>` by `\do@check@aux`, which it can later control the timing.

Now we will apply the same changes, but this time without redefining `\enddocument`: we will instead replace tokens on-the-fly, when `\enddocument` is expanded. This will grant us a slightly safer approach that won't depend so much on the internals of `\enddocument`.

This entire patch should work with the previous definition of `\enddocument` as well (except it cannot be used in the hook), but for now leave previous versions untouched.

The entire patching will reside in the `enddocument` hook:

```
\rvtx@ifformat@geq{2020-10-01}{%
  \AddToHook{enddocument}{\rvtx@enddocument@patch{}}%
}{}
```

This macro will be executed after `\enddocument` has expanded, so all its tokens are now exposed. Here we will assume that `\enddocument` contains the tokens `\@checkend{document}` and `\endgroup`, and use them as delimiters:

```
\protected\long\def\rvtx@enddocument@patch#1#2\@checkend#3{%
  \begingroup
    \edef\x{\detokenize{#3}}%
    \edef\y{\detokenize{document}}%
  \expandafter\endgroup
  \ifx\x\y
    \expandafter\rvtx@enddocument@patch@end
  \else
    \expandafter\rvtx@enddocument@patch@more
  \fi
    {#1#2}{#3}}
\def\rvtx@enddocument@patch@more#1#2{%
  \rvtx@enddocument@patch{#1\@checkend{#2}}}
```

When the `\@checkend{document}` is reached, use `\clearpage` and `\enddocument` as delimiters for the `<read main .aux and final checks>` part, and save it in `\do@check@aux`:

```
\long\def\rvtx@enddocument@patch@end#1#2\clearpage#3\endgroup{%
  \def\do@check@aux{#3\endgroup}%
```

Then execute the code consumed in the previous step:

```
  #1%
  \@checkend{#2}%
```

Do `\clear@document` instead of `\clearpage` and `\check@aux` instead of the code grabbed.

```
  \clear@document
  \check@aux}
\def\check@aux{\do@check@aux}%
```

`\clear@document` The procedure `\clear@document` is responsible for flushing out the last page of the document, if not already done. The procedure then executes those procedures that must wait for execution until after the last page is shipped out. Clients of `ltxutil`, such as `ltxgrid` and `revtex4` will queue these procedures up via `\AfterLastShipout`, if it exists, otherwise by doing `\appdef\clear@document`.

The command `\Call@AfterLastShipout` is provided by Heiko Oberdiek's `atveryend` package. This package is compatible with `ltxutil`.

Note on compatibility with `atveryend`: we arrange for `\Call@AfterLastShipout` to be called from the safety of the output routine, thereby ensuring that all of the procedures queued up by that package's `\AfterLastShipout` are executed at the right time. We also ensure that `\Call@AfterLastShipout` has a default definition, in case the package was never loaded.

```
\def\clear@document{%
 \clearpage
 \do@output@cclv{%
  \Call@AfterLastShipout
 }%
}%
\appdef\class@documenthook{%
 \providecommand\Call@AfterLastShipout{}%
}%
```

6.7 Class Extensions

The LaTeX procedure `\@onefilewithoptions` is the vehicle for reading in a LaTeX class or package. The APS RevTeX class implements the use of what are called "substyles", actually extensions to the class itself. Any document class can do likewise.

`\class@extension` `\class@extensionfile` `\class@ext@hook` A procedure similar to LaTeX's `\@onefilewithoptions`, but as an extension to the current document class.

Read in the given file as if it were a document class file. Usage: `\class@extensionfile {`⟨*class*⟩`} \@extension`, where ⟨*class*⟩ is a file (similar to `aps.rtx`) and where

`\@extension` is the file extension for ⟨*class*⟩. For instance, to read in the file `aps.rtx`, do `\class@extensionfile {aps} \substyle@ext`, where the latter has been define to expand to `.rtx`.

Features supported include passing existing class options on to the class extension, `\AtEndOfClass` processing, a stack that restores `\@currname`, `\@currext`, `\@clsextension`, and the `\catcode` of '@', fall-back to a control sequence name (with leading 'rtx@') if no file exists.

Note that `\LoadClass` gives one the ability to write a class that calls in another class as a (sort of) module: this scheme is like `\LoadClass`, but turned inside out.

```
\def\class@extension#1#2{%
 \IfFileExists{#1.#2}{%
  \expandafter\class@extensionfile\csname ver@\@currname.\@currext\endcsname{#1}#2%
 }{%
  \csname rtx@#1\endcsname
 }%
}%
\def\class@extensionfile#1#2#3{%
 \@pass@ptions#3\@unusedoptionlist{#2}%
 \global\let\@unusedoptionlist\@empty
 \expandafter\class@ext@hook\csname#2.#3-h@@k\endcsname#1{#2}#3%
}%
\def\class@ext@hook#1#2#3#4{%
 \@pushfilename@ltx
 \makeatletter
 \let\CurrentOption\@empty
 \@reset@ptions
 \let#1\@empty
 \xdef\@currname{#3}%
 \global\let\@currext#4%
 \global\let\@clsextension\@currext
 \input{#3.#4}%
 \@ifl@ter#4{#3}#2{%
  \class@info{Class extension later than: #2}%
 }{%
  \class@info{Class extension earlier: #2}%
  \@@end
 }%
 #1%
 \let#1\@undefined
 \expandafter\@p@pfilename@ltx\@currnamestack@ltx\@nil
 \@reset@ptions
}%
```

`\@pushfilename`
`\@p@pfilename`

But! LaTeX does not provide for a class extension other than `.cls`, therefore we must extend LaTeX's file name stack with the file extension of a class extension. This way, procedures like `\ProvidesPackage`, `\OptionNotUsed`, `\ProcessOptions`, `\@reset@ptions` will still work properly.

```
\def\@pushfilename@ltx{%
```

```
\xdef\@currnamestack@ltx{%
 {\@currname}%
 {\@currext}%
 {\@clsextension}%
 {\the\catcode`\@}%
 \@currnamestack@ltx
}%
}%
\def\@p@pfilename@ltx#1#2#3#4#5\@nil{%
 \gdef\@currname{#1}%
 \gdef\@currext{#2}%
 \gdef\@clsextension{#3}%
 \catcode`\@#4\relax
 \gdef\@currnamestack@ltx{#5}%
}%
\global\let\@currnamestack@ltx\@empty
```

We carefully patch LaTeX so that the current value of `\@clsextension` can be restored after reading in a class file.

6.8 Type Tools

`\flushing` Undoes `\centering`. Should also undo `\raggedleft` and `\raggedright`.

```
\def\flushing{%
  \let\\\@normalcr
  \leftskip\z@skip
  \rightskip\z@skip
  \@rightskip\z@skip
  \parfillskip\@flushglue
}%
```

`\@centercr` The `\@centercr` command is the replacement for `\@normalcr` when setting type centered or ragged. Normally, the meaning of `\\` is `\@normalcr`, which LaTeX defines via `\DeclareRobustCommand`. In centered or ragged typesetting, the meaning of `\\` is `\@centercr`, therefore it ought to to be defined via `\DeclareRobustCommand` (but unfortunately is not). The fact that it is not is yet another of LaTeX's early failures that will never get fixed.

The following exemplar fails under LaTeX version 2005/12/01, package `textcase` 2004/10/07 v0.07:

```
%\documentclass{article}%
%\usepackage[overload]{textcase}
%\begin{document}
%\centering
%\section{\MakeTextUppercase{Section\\title}}
%Text
%\end{document}
%
```

The solution is to promote `\@centercr` to a robust command, just the same as `\\`. We do that here without needing to know the meaning of the command.

```
\expandafter\DeclareRobustCommand\expandafter\@centercr\expandafter{\@centercr}%
```

6.9 Display Math

`\eqnarray@LaTeX`
`\eqnarray@fleqn@fixed`

Team LaTeX has stated they will never repair Leslie's broken definition of `eqnarray`. Let us be bold....

Note on `hyperref` package compatibility: that package overrides `\eqnarray` by wrapping it up in a larger procedure, so its changes are compatible with this package's changes.

```
\def\eqnarray@LaTeX{%
   \stepcounter{equation}%
   \def\@currentlabel{\p@equation\theequation}%
   \global\@eqnswtrue
   \m@th
   \global\@eqcnt\z@
   \tabskip\@centering
   \let\\\@eqncr
   $$\everycr{}\halign to\displaywidth\bgroup
       \hskip\@centering$\displaystyle\tabskip\z@skip{##}$\@eqnsel
      &\global\@eqcnt\@ne\hskip \tw@\arraycolsep \hfil${##}$\hfil
      &\global\@eqcnt\tw@ \hskip \tw@\arraycolsep
         $\displaystyle{##}$\hfil\tabskip\@centering
      &\global\@eqcnt\thr@@ \hb@xt@\z@\bgroup\hss##\egroup
         \tabskip\z@skip
      \cr
}
\long\def\eqnarray@fleqn@fixed{%
 \stepcounter{equation}\def\@currentlabel{\p@equation\theequation}%
 \global\@eqnswtrue\m@th\global\@eqcnt\z@
 \tabskip\ltx@mathindent
 \let\\=\@eqncr
 \setlength\abovedisplayskip{\topsep}%
 \ifvmode\addtolength\abovedisplayskip{\partopsep}\fi
 \addtolength\abovedisplayskip{\parskip}%
 \setlength\belowdisplayskip{\abovedisplayskip}%
 \setlength\belowdisplayshortskip{\abovedisplayskip}%
 \setlength\abovedisplayshortskip{\abovedisplayskip}%
 $$%
 \everycr{}%
 \halignt@\linewidth\bgroup
  \hskip\@centering$\displaystyle\tabskip\z@skip{##}$\@eqnsel
  &\global\@eqcnt\@ne
   \hskip\tw@\eqncolsep
   \hfil${{}##{}}$\hfil
  &\global\@eqcnt\tw@
   \hskip\tw@\eqncolsep
```

```
    $\displaystyle{##}$\hfil\tabskip\@centering
   &\global\@eqcnt\thr@@\hb@xt@\z@\bgroup\hss##\egroup
    \tabskip\z@skip
   \cr
}%
\@ifx{\eqnarray\eqnarray@LaTeX}{%
 \class@info{Repairing broken LaTeX eqnarray}%
 \let\eqnarray\eqnarray@fleqn@fixed
 \newlength\eqncolsep
 \setlength\eqncolsep\z@
 \let\eqnarray@LaTeX\relax
 \let\eqnarray@fleqn@fixed\relax
}{}%
```

The macro `\ltx@mathindent` is assigned to the `\tabskip` glue just before the alignment preamble is expanded, the value therefore applying at the left of the first column.

The below value specifies the display math to be set centered, as is common practice. Alternatively, `\tabskip` can be set to a different glue value, accomplishing flush-left display math.

Note that the ltxutil.dtxfleqn.clo package provides its own meaning for the eqnarray environment, which is also broken. We do not patch that package, however.

Bug note: The ltxutil.dtxlineno.sty package detects ltxutil.dtxfleqn.clo by testing whether `\mathindent` is defined, instead of using correct LaTeX 2_ε means. Even though our eqnarray environment is modelled after ltxutil.dtxfleqn.clo, we must program defensively here.

```
\def\ltx@mathindent{\@centering}%
\def\set@eqnarray@skips{}%
```

`\prep@math` `\prep@math@patch`

If we are in vertical mode when display math mode is entered (via `$$`), TeX will first enter horizontal mode, then display math mode; this results in a phantom paragraph containing a single `\hbox` consisting of the `\parindent` box followed by the `\parskipfillskip` glue. Of course, that `\hbox` is accompanied by `\parskip` glue and `\baselineskip` glue.

The `\prep@math` procedure removes the `\parindent` box, thereby (magically) eliminating the phantom paragraph. The `\prep@math@patch` procedure head-patches the `equation` and `eqnarray` environments to accomplish this removal of the phantom paragraph.

Note that there are three remaining ways to enter display math mode that we do not treat: the `displaymath` environment (equivalent to `\[`/`\]`), and the primitive the `$$` markup. I refrain from treating the first case because `displaymath` already detects the case where it is entered from vertical mode: I do not wish to engage in the dubious enterprise of attempting to correct a procedure that is ill conceived from the outset. As to the primitive `$$`, there is no help for users who insist upon employing procedural markup in their documents. in their documents.

```
\def\prep@math{%
 \@ifvmode{\everypar{{\setbox\z@\lastbox}}}{}%
```

```
}%
\def\prep@math@patch{%
 \prepdef\equation{\prep@math}%
 \prepdef\eqnarray{\prep@math}%
}%
```

A document class may invoke `\prep@math@patch` at any point it wishes to prevent the appearance of the phantom paragraph: it may be a global declaration or a local one.

We fail to patch `\[`, `\equation`, however.

6.10 Footnotes

`\footnotemark` `\footnotetest` `\ltx@xfootnote` `\ltx@footmark` `\ltx@foottext` `\ltx@make@current@footnote`

We repair an error in the LaTeX kernel (see `ltfloat.dtx`) involving footnotes. The symptom is that the `\footnotemark` command does not work properly within a `minipage` environment. The source of the problem is in the way the `\footnotemark` and `\@xfootnotemark` procedures are defined: they do not share the method, used by `\footnote` and other procedures, that allows a context switch to change the way footnotes behave within a minipage environment. This is a LaTeX bug of long standing; our fix dates to 1987.

While we are at it, we rewrite both the `\footnote`, `\footnotemark` and `\footnotetext` procedures, achieving a cleaner separation of syntax and semantics. Note that the `\@footnotetext` procedure is not involved in context switching; `hyperref` will take over that procedure, substituting its own processing around its argument and passing this to `\H@@footnotetext`. We anticipate this, and do our context switching on `\H@@footnotetext`.

The `\@makefnmark` continues as the method of formatting the footnote mark.

A note about the context switch mentioned above: the `minipage` environment executes the following in order to alter the way footnotes behave:

```
%\def\@mpfn{mpfootnote}%
%\def\thempfn{\thempfootnote}%
%\let\@footnotetext\@mpfootnotetext
%\let\@makefnmark\@mpmakefnmark
%\c@mpfootnote\z@
```

This code changes the counter used in autonumbered footnotes, the choice of footnote marker, and the procedure used on the footnote text. Changing the counter is needed because minipage footnotes are in their own sequence, and the footnote marker is customarily different within a minipage. The procedure that works on the footnote text must be different because the footnotes are placed at the bottom of the minipage, not the bottom of the text column.

Note that LaTeX initially defines `\@mpfn` as `footnote` and `\thempfn` as `\thefootnote`, so we are initially doing general footnotes.

Any procedure that establishes a minipage-like context (e.g., floats) can do the same as the mimipage context switch illustrated above.

Three user-level command, `\footnote`, `\footnotemark`, and `\footnotetext` are defined (see the LaTeX manual for user-level details).

\footnote The first user-level command is \footnote. A simple way to look at this command is to think of it as \footnotemark [⟨*number*⟩] \footnotetext [⟨*number*⟩] {⟨*text*⟩}, where the optional argument is the same in both calls. We also define a syntactical helper procedure \ltx@xfootnote.

We employ the procedures \ltx@stp@footproc and \ltx@def@footproc, passing in the procedure to execute, in this case \ltx@footmark, which sets the footnote mark. In any case, we end on the procedure \ltx@foottext, which sets the footnote text.

```
\def\footnote{\@ifnextchar[\ltx@xfootnote\ltx@yfootnote}%
\def\ltx@xfootnote[#1]{%
 \ltx@def@footproc\ltx@footmark[#1]%
 \expandafter\ltx@foottext\expandafter{\the\csname c@\@mpfn\endcsname}%
}%
\def\ltx@yfootnote{%
 \ltx@stp@footproc\ltx@footmark
 \expandafter\ltx@foottext\expandafter{\the\csname c@\@mpfn\endcsname}%
}%
```

The \footmark user-level command is next. Here we use the procedures \ltx@stp@footproc and \ltx@def@footproc again, but unlike \footnote, we do not set the footnote text.

```
\def\footnotemark{\@ifnextchar[\ltx@xfootmark\ltx@yfootmark}%
\def\ltx@xfootmark{\ltx@def@footproc\ltx@footmark}%
\def\ltx@yfootmark{\ltx@stp@footproc\ltx@footmark}%
\def\ltx@footmark#1{%
 \leavevmode
 \ifhmode\edef\@x@sf{\the\spacefactor}\nobreak\fi
 \begingroup
  \expandafter\ltx@make@current@footnote\expandafter{\@mpfn}{#1}%
  \expandafter\@argswap@val\expandafter{\Hy@footnote@currentHref}{\hyper@linkstart {link}}%
   \@makefnmark
  \hyper@linkend
 \endgroup
 \ifhmode\spacefactor\@x@sf\fi
 \relax
}%
```

The third user-level command is \footnotetext. As with \footnotemark, we use the procedures \ltx@stp@footproc and \ltx@def@footproc, this time passing in the procedure \ltx@foottext, which sets the footnote text.

```
\def\footnotetext{\@ifnextchar[\ltx@xfoottext\ltx@yfoottext}%
\def\ltx@xfoottext{\ltx@def@footproc\ltx@foottext}%
\def\ltx@yfoottext{\ltx@stp@footproc\ltx@foottext}%
\long\def\ltx@foottext#1#2{%
 \begingroup
  \expandafter\ltx@make@current@footnote\expandafter{\@mpfn}{#1}%
  \@footnotetext{#2}%
 \endgroup
}%
```

Here are the definitions of the procedures `\ltx@stp@footproc` and `\ltx@def@footproc`. The require argument is the procedure to execute afterwards, and `\ltx@def@footproc` parses a bracket-delimited argument (it is not optional). In each case the given procedure is executed with an argument prepared for it: the value of the footnote counter.

```
\def\ltx@def@footproc#1[#2]{%
 \begingroup
   \csname c@\@mpfn\endcsname #2\relax
   \unrestored@protected@xdef\@thefnmark{\thempfn}%
 \expandafter\endgroup
 \expandafter#1%
 \expandafter{\the\csname c@\@mpfn\endcsname}%
}%
\def\ltx@stp@footproc#1{%
 \expandafter\stepcounter\expandafter{\@mpfn}%
 \protected@xdef\@thefnmark{\thempfn}%
 \expandafter#1%
 \expandafter{\the\csname c@\@mpfn\endcsname}%
}%
```

Here we provide for our good friend `hyperref` to enter in like a bull in a china shop. If it is not loaded, we do what it would have done, but gentlier and without hypertext functionality.

```
\appdef\class@documenthook{%
 \let\footnote@latex\footnote
 \@ifpackageloaded{hyperref}{}{%
  \let\H@@footnotetext\@footnotetext
  \def\@footnotetext{\H@@footnotetext}%
  \let\H@@mpfootnotetext\@mpfootnotetext
  \def\@mpfootnotetext{\H@@mpfootnotetext}%
 }%
}%
```

In the following, we must use LaTeX's rococco equipment in the form of `\protected@edef`, because of the presence of a font switch in the meaning of `\thempfootnote`. But, really, isn't this a sloppy conflation of semantics and presentation?

```
\def\ltx@make@current@footnote#1#2{%
  \csname c@#1\endcsname#2\relax
  \protected@edef\Hy@footnote@currentHref{\@currentHref-#1.\csname the#1\endcsname}%
}%
\def\thempfootnote@latex{{\itshape \@alph \c@mpfootnote }}%
\def\ltx@thempfootnote{\@alph\c@mpfootnote}%
\@ifx{\thempfootnote\thempfootnote@latex}{%
 \class@info{Repairing hyperref-unfriendly LaTeX definition of \string\mpfootnote}%
 \let\thempfootnote\ltx@thempfootnote
}{}%
```

Note on `hyperref` compatibility: In its "Automated LaTeX hypertext cross-references", the `hyperref` package alters footnote processing, but it does nothing

to address the several issues of concern to us.

The `hyperref` package takes over the `\@mpfootnotetext` and `\@footnotetext` procedures, wrapping the argument in its own code. It also rewrites `\@footnotemark`, making it a hyperlink, and `\@xfootnotenext`, removing from it all hypertext capabilities.

However, if the `\footnotemark` command has been supplied with an optional argument, `hyperref`'s changes do not apply: it punts in this case.

At the same time, it attempts to turn off its changes during `\maketitle` processing, destroying one of the capabilities we desire.

We make ourself `hyperref` savvy: we re-implement footnote processing, using `hyperref` capabilities if that package has been loaded.

Any other package that rewrites LaTeX's footnote macros will be incompatible with this package.

Two thoughts about `hyperref`: what for does it define `\realfootnote`? Apparently even SR himself cannot remember.

Also: a document class that desires high hypertext capabilities might well wish to reimplement `\maketitle` so that footnotes called out from there are hypertext links: the `hyperref` package's "Automated LaTeX hypertext cross-references" does not do any of this:

> But the special footnotes in `\maketitle` are much too hard to deal with properly. Let them revert to plain behaviour.

Note that the document class, in reimplementing `\maketitle`, must ensure that the `hyperref` package does not clobber its own definition!

`\@footnotetext`
`\@mpfootnotetext`
`\@tpfootnotetext`
`\make@footnotetext`
`\set@footnotewidth`

The two procedures `\@footnotetext` and `\@mpfootnotetext` share code. We make that explicit here.

Note that the procedure calling `\make@footnotetext` will open a group with `\bgroup` which is then closed by `\minipagefootnote@drop`.

Difference from LaTeX: here we do not set `\floatingpenalty` to infinity. Doing this must date back to a time when LaTeX could not accomodate split insertions (footnotes). I cannot think of any other reason to do have done this. At any rate, with the `ltxgrid` package, split insertions are properly taken care of, so we allow it.

We provide the hook `\set@footnotewidth` that sets the footnote on a particular measure. Some page grids are such as to set a footnote in a context where `\columnwidth`is not the right parameter to use for the set width of a footnote. In such a case, for the applicable scope, you should define `\set@footnotewidth` to perform this job correctly.

If we are setting type on multiple page grids, we must still ensure that all footnotes that find their way into the `\footins` insert register are set on the same width. This implies the need for a document to have an "overall" page grid, which determines the set width of all footnotes with the exception of minipage footnotes.

In general, remember that footnotes, like all insertions (including floats), are a step outside of the galley context, and all aspects of insertions need to be properly handled, including the set width.

```
\def\@makefnmark{%
 \hbox{%
  \@textsuperscript{%
   \normalfont\itshape\@thefnmark
  }%
 }%
}%
\long\def\@footnotetext{%
 \insert\footins\bgroup
  \make@footnotetext
}%
\long\def\@mpfootnotetext{%
 \minipagefootnote@pick
  \make@footnotetext
}%
```

Procedure `\make@footnotetext` sets the footnote `#1` into type, with the proper font, color, leading, width, and label in effect. It also establishes a strut and null glue at the end of the last paragraph of the footnote; The strut helps compensate for the lack of `\interlineskip` glue between `\insert`s; the glue establishes a feasible `\vsplit` point between footnotes.

Note that in the title block (`ltxfront`), the alternative definition, under the name `\frontmatter@footnotetext`, is used. The only material difference there is the reference to `\frontmatter@makefntext` instead of `\@makefntext`.

Dependency note: the `\@makefntext` procedure is used to further process the footnote text and to execute the `\@makefnmark` procedure to produce the footnote mark. The definition of the former is customarily found in the document class (hereunder that of ltxutil.dtxarticle.cls), the latter in ltxutil.dtxlatex.ltx. They are as follows:

```
%\newcommand\@makefntext[1]{%
% \parindent 1em\noindent
% \hb@xt@1.8em{\hss\@makefnmark}%
% #1%
%}%
%\def\@makefnmark{%
% \hbox{\@textsuperscript{\normalfont\@thefnmark}}%
%}%
%
```

```
\long\def\make@footnotetext#1{%
  \set@footnotefont
```

As noted above, we do *not* do `\floatingpenalty` `\@MM`, as in standard LaTeX.

```
  \set@footnotewidth
  \@parboxrestore
  \protected@edef\@currentlabel{%
```

Note that we employ `\@mpfn` as a level of redirection for the footnotecounter.

```
643    \csname p@\@mpfn\endcsname\@thefnmark
644   }%
645   \color@begingroup
646    \@makefntext{%
647     \rule\z@\footnotesep\ignorespaces#1%
```

The following strut and glue are for spacing and splitting, as mentioned above.

```
648     \@finalstrut\strutbox\vadjust{\vskip\z@skip}%
649    }%
650   \color@endgroup
651  \minipagefootnote@drop
652 }%
```

`\set@footnotefont` is the procedure for setting the font of a footnote. Other aspects of the environment may be set using this hook.

```
653 \def\set@footnotefont{%
654   \reset@font\footnotesize
655   \interlinepenalty\interfootnotelinepenalty
656   \splittopskip\footnotesep
657   \splitmaxdepth\dp\strutbox
658 }%
```

`\set@footnotewidth` is the procedure for setting the width of a footnote. The default page grid, a single, full-width column, sets footnotes on the width of the text.

```
659 \def\set@footnotewidth{\set@footnotewidth@one}%
```

6.11 Floats

6.11.1 Usage notes

We extend the LaTeX kernel for three purposes:

1. When the `\footnote` command is used within the scope of a float, we do as `minipage` does.
2. We provide a mechanism to write floats out to an external stream for temporary storage (deferred floats).
3. We provide mechanism for placing a float `here` invariably, that is, floats are unfloated. This mechanism is used to read the external stream mentioned above.

To use these mechanisms, the document class should define a float, say, `figure` as per usual, and in addition:

1. Optionally define an alternative, say `figure@write` as follows:

```
\newenvironment{figure@write}{%
% \write@float{figure}%
```

```
%}{%
% \endwrite@float
%}
```

That is, the alternative environment executes `\write@float` instead of `\@float`. Note that this step is not needed if the float environment is defined in the simple way of `classes.dtx`. However, an environment like `longtable` will require it.

2. Install into `\AtBeginDocument` a call to `\do@if@floats`, with the float name and an appropriate file extension as its arguments.

   ```
   \appdef\class@documenthook{\do@if@floats{figure}{.fgx}}
   ```

3. Optionally define a text entity `\figuresname` that will be the text of the head that is set over the deferred floats. If not defined, there will be no head.

4. Optionally define a user-level command to allow the document to determine where the figures are printed out (default is to print at end of document). E.g.,

   ```
   \newcommand\printfigures{\print@float{figure}}
   ```

5. Install into `\appdef\class@enddocumenthook` a call to `\printfigures`, or, if the latter is not defined, as follows:

   ```
   \appdef\class@enddocumenthook{\print@float{figure}}
   ```

   Note that installing this command into `\AtBeginDocument`is best done earlier than calls that assume the last page of the document is at hand.

6.11.2 Robustifying fragile commands

Certain of LaTeX's commands cannot be written out to a file or appear within a `\mark` command argument because they do calculations during expansion. We provide for a little help, but without changing the meanings of these commands.

`\addtocontents`
`\robustify@contents`

```
\def\robustify@contents{%
 \let \label \@gobble
 \let \index \@gobble
 \let \glossary \@gobble
 \let\footnote \@gobble
 \def\({\string\(}%
 \def\){\string\)}%
 \def\\{\string\\}%
}%
\long\def\addtocontents#1#2{%
 \protected@write\@auxout{\robustify@contents}{\string \@writefile {#1}{#2}}%
}%
```

6.11.3 Preparing for the `hyperref` package

`\addcontentsline` `\label` `\ltx@contentsline`

The `hyperref` package assumes that the `\contentsline` command will be given four arguments. Therefore it cannot successfully process a ltxutil.dtx.toc file that had been written by standard LaTeX. We fix things up by always writing that fourth argument and by supplying a `\contentsline` command that can read them.

We also give the `\newlabel` command's second argument five tokens.

Finally, we wrap LaTeX's `\contentsline` command with code to detect the case where the expected procedure is not defined, and we give it a syntax with no semantics.

We switch over to this new definition only after `hyperref` has loaded.

```
\def\addcontentsline#1#2#3{%
 \addtocontents{#1}{%
  \protect\contentsline{#2}{#3}{\thepage}{}%
 }%
}%
\def\label#1{%
 \@bsphack
  \protected@write\@auxout{}{%
   \string\newlabel{#1}{{\@currentlabel}{\thepage}{}{}{}}%
  }%
 \@esphack
}%
\def\ltx@contentsline#1{%
 \expandafter\@ifnotrelax\csname l@#1\endcsname{}{%
  \expandafter\let\csname l@#1\endcsname\@gobbletwo
 }%
 \contentsline@latex{#1}%
}%
\appdef\document@inithook{%
 \let\contentsline@latex\contentsline
 \let\contentsline\ltx@contentsline
}%
```

6.11.4 Footnotes within floats, unfloating floats, float font

`\caption`

DPC: Er a bit of a hack, but seems best way of supporting normal LaTeX syntax at this point: If a caption is used below a table, then put out the footnotes before the caption.

```
\appdef\class@documenthook{%
 \prepdef\caption{\minipagefootnote@here}%
}%
```

Note on `hyperref` compatibility: this change to the `\caption` command is compatible with the "Automated LaTeX hypertext cross-references" patches of that package.

All the same, I think Sebastian's changes to `\caption` and `\@caption` could bear with some improvement. The following implementation requires knowing only the pattern part of the `\@caption` macro:

```
%\def\caption{%
%  \H@refstepcounter\@captype
%  \hyper@makecurrent{\@captype}%
%  \@dblarg{\H@caption\@captype}%
%}%
%\def\H@caption#1[#2]#3{%
% \@caption{#1}[#2]{%
%  \ifHy@nesting
%   \hyper@@anchor{\@currentHref}{#3}%
%  \else
%   \hyper@@anchor{\@currentHref}{\relax}#3%
%  \fi
% }%
%}
```

\minipagefootnote@init
\minipagefootnote@here
\minipagefootnote@foot
\minipagefootnote@pick
\minipagefootnote@drop

Procedure to deal with footnotes accumulated within a minipage environment. These procedures encapsulate all uses of the `\@mpfootins` box.

Note: `\minipagefootnote@here` must *not* be executed within the MVL!

```
\def\minipagefootnote@init{%
 \setbox\@mpfootins\box\voidb@x
}%
\def\minipagefootnote@pick{%
 \global\setbox\@mpfootins\vbox\bgroup
  \unvbox\@mpfootins
}%
\def\minipagefootnote@drop{%
 \egroup
}%
\def\minipagefootnote@here{%
    \par
    \@ifvoid\@mpfootins{}{%
      \vskip\skip\@mpfootins
      \fullinterlineskip
      \@ifinner{%
       \vtop{\unvcopy\@mpfootins}%
       {\setbox\z@\lastbox}%
      }{}%
      \unvbox\@mpfootins
    }%
}%
\def\minipagefootnote@foot{%
 \@ifvoid\@mpfootins{}{%
  \insert\footins\bgroup\unvbox\@mpfootins\egroup
 }%
}%
\def\endminipage{%
    \par
    \unskip
    \minipagefootnote@here
    \@minipagefalse   %% added 24 May 89
```

```
  \color@endgroup
  \egroup
  \expandafter\@iiiparbox\@mpargs{\unvbox\@tempboxa}%
}%
```

`\floats@sw` The Boolean `\floats@sw` signifies that floats are to be floated; if false, that floats are to be deferred to the end of the document. Note that the assignment of this Boolean is to be overridden by the document class in response to user-selected options.

```
\@booleantrue\floats@sw
```

`\@xfloat` `\@mpmakefntext` The float start-code is redefined to set up footnotes in the style of minipage. Also, the `\floats@sw` Boolean informs us that floats are to be all placed `here`. Note that, to protect against the Boolean being undefined at this late hour, we default it globally to true.

```
\let\@xfloat@LaTeX\@xfloat
\def\@xfloat#1[#2]{%
  \@xfloat@prep
  \@nameuse{fp@proc@#2}%
  \floats@sw{\@xfloat@LaTeX{#1}[#2]}{\@xfloat@anchored{#1}[]}%
}%
\def\@xfloat@prep{%
  \ltx@footnote@pop
  \def\@mpfn{mpfootnote}%
  \def\thempfn{\thempfootnote}%
  \c@mpfootnote\z@
  \let\H@@footnotetext\H@@mpfootnotetext
}%
\let\ltx@footnote@pop\@empty
\def\@xfloat@anchored#1[#2]{%
 \def\@captype{#1}%
 \begin@float@pagebreak
  \let\end@float\end@float@anchored
  \let\end@dblfloat\end@float@anchored
        \hsize\columnwidth
        \@parboxrestore
        \@floatboxreset
  \minipagefootnote@init
}%
\def\end@float@anchored{%
  \minipagefootnote@here
  \par\vskip\z@skip
 \par
 \end@float@pagebreak
}%
\def\begin@float@pagebreak{\par\addvspace\intextsep}%
\def\end@float@pagebreak{\par\addvspace\intextsep}%
\def\@mpmakefntext#1{%
 \parindent=1em
```

```
\noindent
\hb@xt@1em{\hss\@makefnmark}%
#1%
}%
```

6.11.5 Writing floats out to a file

`\do@if@floats` The procedure `\do@if@floats` should be executed at `\class@documenthook` time: it arranges to write out the floats of the given class to a temporary file, to be read back later (deferred floats), given that `\floats@sw` is false. Note that, to protect against the Boolean being undefined at this late hour, we default it globally to true.

```
\def\do@if@floats#1#2{%
 \floats@sw{}{%
```

Open the stream to save out the document's floats of this class.

```
  \expandafter\newwrite
              \csname#1write\endcsname
  \expandafter\def
              \csname#1@stream\endcsname{\jobname#2}%
  \expandafter\immediate
  \expandafter\openout
              \csname#1write\endcsname
              \csname#1@stream\endcsname\relax
```

Swap environments. If the class writer has defined, e.g., `figure@write`, then we use this as the procedure to execute for writing the float out to the external stream. Otherwise, the replacement of `\@float` by `\write@float` should do the right thing for float environments defined in the simple way of `classes.dtx`.

```
  \@ifxundefined\@float@LaTeX{%
   \let\@float@LaTeX\@float
   \let\@dblfloat@LaTeX\@dblfloat
   \let\@float\write@float
   \let\@dblfloat\write@floats
  }{}%
  \let@environment{#1@float}{#1}%
  \let@environment{#1@floats}{#1*}%
  \@ifxundefined@cs{#1@write}{}{%
   \let@environment{#1}{#1@write}%
  }%
 }%
}%
```

`\print@float` The procedure `\print@float` prints out the deferred floats.

Here, we make use of the `\floats@sw` Boolean to select the non-floating type of processing.

```
\def\triggerpar{\leavevmode\@@par}%
\def\oneapage{\def\begin@float@pagebreak{\newpage}\def\end@float@pagebreak{\newpage}}%
\def\print@float#1#2{%
```

```
\lengthcheck@sw{%
 \total@float{#1}%
}{}%
\@ifxundefined@cs{#1write}{}{%
 \begingroup
  \@booleanfalse\floats@sw
  #2%
  \raggedbottom
  \def\array@default{v}% floats must
  \let\@float\@float@LaTeX
  \let\@dblfloat\@dblfloat@LaTeX
  \let\trigger@float@par\triggerpar
  \let@environment{#1}{#1@float}%
  \let@environment{#1*}{#1@floats}%
  \expandafter\prepdef\csname#1\endcsname{\trigger@float@par}%
  \expandafter\prepdef\csname#1*\endcsname{\trigger@float@par}%
  \@namedef{fps@#1}{h!}%
  \expandafter\immediate
  \expandafter\closeout
              \csname#1write\endcsname
  \everypar{%
   \global\let\trigger@float@par\relax
   \global\everypar{}\setbox\z@\lastbox
   \@ifxundefined@cs{#1sname}{}{%
    \begin@float@pagebreak
    \expandafter\section
    \expandafter*%
    \expandafter{%
                 \csname#1sname\endcsname
                }%
   }%
  }%
  \input{\csname#1@stream\endcsname}%
 \endgroup
 \global\expandafter\let\csname#1write\endcsname\relax
}%
}%
```

`\tally@float` `\total@float`

If we are tallying column inches, `\tally@float` tallies a contribution to `\ftype@\@captype`, depending upon the width of `\@currbox`. In effect, each float class is tallied in two sections, one for narrow, one for wide floats.

If statistics are wanted, `\total@float` logs the tally for the given float class. The quantity `\@twopowerfourteen` is 2^{14}, `\@twopowertwo` is 2^2.

```
\chardef\@xvi=16\relax
\mathchardef\@twopowerfourteen="4000
\mathchardef\@twopowertwo="4
\def\tally@float#1{%
 \begingroup
```

We strip all but the least significant 5 bits from `\count \@currbox`, and put them

into \@tempcnta. We then subtract 16 from \count \@currbox(unless this would make it negative), effectively reversing the process carried out in \@float.

```
  \@tempcnta\count\@currbox
  \divide\@tempcnta\@xxxii
  \multiply\@tempcnta\@xxxii
  \advance\count\@currbox-\@tempcnta
  \divide\@tempcnta\@xxxii
  \@ifnum{\count\@currbox>\@xvi}{%
   \advance\count\@currbox-\@xvi\@booleantrue\@temp@sw
  }{%
   \@booleanfalse\@temp@sw
  }%
```

If so desired, we log the characteristics of this float object: float class and float placement parameters, height, depth, and width.

```
  \show@box@size@sw{%
   \class@info{Float #1
    (\the\@tempcnta)[\@temp@sw{16+}{}\the\count\@currbox]^^J%
    (\the\ht\@currbox+\the\dp\@currbox)X\the\wd\@currbox
   }%
  }{}%
 \endgroup
```

Here we tally the height of this float object.

```
 \expandafter\let
 \expandafter\@tempa
             \csname fbox@\csname ftype@#1\endcsname\endcsname
 \@ifnotrelax\@tempa{%
  \@ifhbox\@tempa{%
   \setbox\@tempboxa\vbox{\unvcopy\@currbox\hrule}%
   \dimen@\ht\@tempboxa
   \divide\dimen@\@twopowerfourteen
   \@ifdim{\wd\@tempboxa<\textwidth}{%
    \advance\dimen@\ht\@tempa
    \global\ht\@tempa\dimen@
   }{%
    \advance\dimen@\dp\@tempa
    \global\dp\@tempa\dimen@
   }%
  }{}%
 }{}%
}%
\def\total@float#1{%
 \expandafter\let
 \expandafter\@tempa
             \csname fbox@\csname ftype@#1\endcsname\endcsname
 \@ifnotrelax\@tempa{%
  \@ifhbox\@tempa{%
   \@tempdima\the\ht\@tempa\divide\@tempdima\@twopowertwo\@tempcnta\@tempdima
   \@tempdimb\the\dp\@tempa\divide\@tempdimb\@twopowertwo\@tempcntb\@tempdimb
```

```
   \class@info{Total #1: Column(\the\@tempcnta pt), Page(\the\@tempcnta pt)}%
  }{}%
 }{}%
}%
```

`\write@float` `\write@floats` `\write@@float`

Handles the case where the name of the float is the same as that of the stream. Note that `longtable` does *not* fit this case. Note also: `\write@float` is *not* a user-level environment, therefore it is properly not defined with `\newenvironment`.

```
\def\write@float#1{\write@@float{#1}{#1}}%
\def\endwrite@float{\@Esphack}%
\def\write@floats#1{\write@@float{#1*}{#1}}%
\def\endwrite@floats{\@Esphack}%
```

`\write@@float`

```
\def\write@@float#1#2{%
  \ifhmode
     \@bsphack
  \fi
  \chardef\@tempc\csname#2write\endcsname
  \toks@{\begin{#1}}%
  \def\@tempb{#1}%
  \expandafter\let\csname end#1\endcsname\endwrite@float
  \catcode`\^^M\active
  \@makeother\{\@makeother\}\@makeother\%
  \write@floatline
}%
```

`\write@floatline` `\@write@floatline` `\float@end@tag`

The procedure `\write@floatline` only parses; it passes its result to `\@write@floatline`, which writes the line to output, then tests the line for the `\end{`⟨*float*⟩`}` tokens with aid of the `\float@end@tag` procedure.

```
\begingroup
 \catcode`\[\the\catcode`\{\catcode`\]\the\catcode`\}\@makeother\{\@makeother\}%
 \gdef\float@end@tag#1\end{#2}#3\@nul[%
  \def\@tempa[#2]%
  \@ifx[\@tempa\@tempb][\end[#2]][\write@floatline]%
 ]%
 \obeylines%
 \gdef\write@floatline#1^^M[%
  \begingroup%
   \newlinechar`\^^M%
   \toks@\expandafter[\the\toks@#1]\immediate\write\@tempc[\the\toks@]%
  \endgroup%
  \toks@[]%
  \float@end@tag#1\end{}\@nul%
 ]%
\endgroup
```

6.12 Counters

The following definitions override those of the LaTeX kernel, providing for a greater range of inputs.

```
\def\@alph#1{\ifcase#1\or a\or b\or c\or d\else\@ialph{#1}\fi}

\def\@ialph#1{\ifcase#1\or \or \or \or \or e\or f\or g\or h\or i\or j\or
  k\or l\or m\or n\or o\or p\or q\or r\or s\or t\or u\or v\or w\or x\or
  y\or z\or aa\or bb\or cc\or dd\or ee\or ff\or gg\or hh\or ii\or jj\or
  kk\or ll\or mm\or nn\or oo\or pp\or qq\or rr\or ss\or tt\or uu\or
  vv\or ww\or xx\or yy\or zz\else\@ctrerr\fi}
```

6.13 Customization of Sections

Patch the standard LaTeX sectioning procedure to:

- Allow a sectioning command to trigger the title page, or more generally to recognize that it is the first object in the document, so we headpatch `\@startsection`.
- Allow a tail command in `#6` to uppercase the title, so we retain DPC's braces.
- Allow each type of sectioning command to format its number differently, so we generalize `\@seccntformat`.
- Allow each type of sectioning command to format its argument differently, so we generalize `\@hangfrom`.
- Allow the starred form of the command to mark (the running head) and make an entry in the TOC, so we put `\@ssect` on the same footing as `\@sect`.

  Note that the tokens passed to the TOC now are *not* the optional argument of the command, but the required. This means that the user can no longer use the former to put variant content in to the TOC as the Manual says.

  Instead, the optional argument is used to put an alternative title into the running headers, a better choice.

`\@startsection` Patch a head hook into the basic sectioning command. Treat `\@sect` and `\@ssect` on an equal footing: now their pattern parts are identical.

```
\def\@startsection#1#2#3#4#5#6{%
 \@startsection@hook
 \if@noskipsec \leavevmode \fi
 \par
 \@tempskipa #4\relax
 \@afterindenttrue
 \ifdim \@tempskipa <\z@
  \@tempskipa -\@tempskipa \@afterindentfalse
 \fi
 \if@nobreak
  \everypar{}%
```

```
 \else
  \addpenalty\@secpenalty\addvspace\@tempskipa
 \fi
 \@ifstar
  {\@dblarg{\@ssect@ltx{#1}{#2}{#3}{#4}{#5}{#6}}}%
  {\@dblarg{\@sect@ltx {#1}{#2}{#3}{#4}{#5}{#6}}}%
}%
\def\@startsection@hook{}%
```

\@sect When defining \@svsec, do not expand \@seccntformat. Put brace characters back where they were before David Carlisle got at them (i.e., as if \@hangfrom had two arguments). Protect the mark mechanism from an undefined meaning. Pass #8 to the TOC instead of #7. Remove \relax from the replacement part of \@svsec.

The procedure \@hangfrom and \@runin@to can be used to process the argument of the head. The head can define, e.g., \@hangfrom@section, to do its own processing.

In using \H@refstepcounter in place of \refstepcounter we rely on either loading before any package that patches the latter, or the convention that the former is the original LaTeX procedure.

```
\class@info{Repairing broken LateX \string\@sect}%
\def\@sect@ltx#1#2#3#4#5#6[#7]#8{%
  \@ifnum{#2>\c@secnumdepth}{%
    \def\H@svsec{\phantomsection}%
    \let\@svsec\@empty
  }{%
    \H@refstepcounter{#1}%
    \def\H@svsec{%
     \phantomsection
    }%
    \protected@edef\@svsec{{#1}}%
    \@ifundefined{@#1cntformat}{%
     \prepdef\@svsec\@seccntformat
    }{%
     \expandafter\prepdef
     \expandafter\@svsec
                 \csname @#1cntformat\endcsname
    }%
  }%
  \@tempskipa #5\relax
  \@ifdim{\@tempskipa>\z@}{%
    \begingroup
      \interlinepenalty \@M
      #6{%
       \@ifundefined{@hangfrom@#1}{\@hang@from}{\csname @hangfrom@#1\endcsname}%
       {\hskip#3\relax\H@svsec}{\@svsec}{#8}%
      }%
      \@@par
    \endgroup
```

```
    \@ifundefined{#1mark}{\@gobble}{\csname #1mark\endcsname}{#7}%
    \addcontentsline{toc}{#1}{%
      \@ifnum{#2>\c@secnumdepth}{%
       \protect\numberline{}%
      }{%
       \protect\numberline{\csname the#1\endcsname}%
      }%
      #8}%
  }{%
    \def\@svsechd{%
      #6{%
       \@ifundefined{@runin@to@#1}{\@runin@to}{\csname @runin@to@#1\endcsname}%
       {\hskip#3\relax\H@svsec}{\@svsec}{#8}%
      }%
      \@ifundefined{#1mark}{\@gobble}{\csname #1mark\endcsname}{#7}%
      \addcontentsline{toc}{#1}{%
        \@ifnum{#2>\c@secnumdepth}{%
         \protect\numberline{}%
        }{%
         \protect\numberline{\csname the#1\endcsname}%
        }%
        #8}%
    }%
  }%
  \@xsect{#5}%
}%
\def\@hang@from#1#2#3{\@hangfrom{#1#2}#3}%
\def\@runin@to #1#2#3{#1#2#3}%
```

`\@ssect` Put brace characters back where they were before David Carlisle got at them (as if `\@hangfrom` has two arguments). Possibly set a mark. Make a TOC entry.

Note that, for compatibility with the `hyperref` package, we need to provide the interface required by that package (actually required by `pdfmark.def` and `nameref.sty`), namely the definition of `\@currentlabelname` (but now removed), the insertion of the procedure `\Sectionformat` (but why is this needed?), and the call to `\phantomsection` (which must precede the call to `\addcontentsline`). We also have to sidestep the patch to `\@ssect` in that same file, therefore we use a different control sequence name in the call from `\@startsection`.

```
\def\@ssect@ltx#1#2#3#4#5#6[#7]#8{%
```

Removed `\def\@currentlabelname{#8}`

```
  \def\H@svsec{\phantomsection}%
  \@tempskipa #5\relax
  \@ifdim{\@tempskipa>\z@}{%
    \begingroup
      \interlinepenalty \@M
      #6{%
       \@ifundefined{@hangfroms@#1}{\@hang@froms}{\csname @hangfroms@#1\endcsname}%
```

Removed `{\hskip#3\relax\H@svsec}{\Sectionformat{#8}{#1}}`

```
      {\hskip#3\relax\H@svsec}{#8}%
    }%
    \@@par
  \endgroup
  \@ifundefined{#1smark}{\@gobble}{\csname #1smark\endcsname}{#7}%
  \addcontentsline{toc}{#1}{\protect\numberline{}#8}%
 }{%
  \def\@svsechd{%
    #6{%
     \@ifundefined{@runin@tos@#1}{\@runin@tos}{\csname @runin@tos@#1\endcsname}%
```

Removed `{\hskip#3\relax\H@svsec}{\Sectionformat{#8}{#1}}`

```
     {\hskip#3\relax\H@svsec}{#8}%
    }%
    \@ifundefined{#1smark}{\@gobble}{\csname #1smark\endcsname}{#7}%
    \addcontentsline{toc}{#1}{\protect\numberline{}#8}%
  }%
 }%
 \@xsect{#5}%
}%
\def\@hang@froms#1#2{#1#2}%
\def\@runin@tos #1#2{#1#2}%
```

`\init@hyperref` Document classes that incorporate this package will be `hyperref`-savvy. (To accomplish this, we ensure that `\hyperanchor` and `\hyper@last` are both defined.) Being `hyperref`-savvy levels some requirements on us, but the benefits are many.

One is that the TOC will not get amnesia and require a full set of three typesetting runs before its formatting is stable. Instead, only two runs are required: the first updates the auxiliary file, the second the TOC. However, the formatting of the document does not change.

Another aspect of being `hyperref`-savvy is that the syntax of commands in the `.aux` file will not change if `hyperref` is turned on or off.

Note that `\hyper@anchorstart` and `\hyper@anchorend` constitute the programming interface for a hypertext anchor (the target of a hypertext link); `\hyper@linkstart` and `\hyper@linkend` are the interface for a hypertext link.

```
\def\init@hyperref{%
 \providecommand\phantomsection{}%
 \providecommand\hyper@makecurrent[1]{}%
 \providecommand\Hy@raisedlink[1]{}%
 \providecommand\hyper@anchorstart[1]{}%
 \providecommand\hyper@anchorend{}%
 \providecommand\hyper@linkstart[2]{}%
 \providecommand\hyper@linkend{}%
 \providecommand\@currentHref{}%
}%
\let\H@refstepcounter\refstepcounter
\appdef\document@inithook{%
 \init@hyperref
}%
```

`\sec@upcase` Upper case for sections (optional upper case items). These are created so that some headings can be toggled between mixed case and upper case readily. Headings that might be changed can be wrapped in the style file in `\sec@upcase{`⟨*text*⟩`}` constructs; the expansion of `\sec@upcase` is controlled here. It is `\relax` by default (mixed case heads), and can easily be changed to `\uppercase` if desired. If mixed-case headings are wanted by the editor, authors *must* supply mixed case text, although this is what authors should be doing anyway. (Mixed can be converted to upper, but the reverse transformation cannot be automated.)

The following setting gives the LaTeX default.

```
\def\sec@upcase#1{\relax{#1}}%
```

6.14 Patch the `tabular` and `array` Environments

`\endtabular` `\endarray` We headpatch the begin processing and tailpatch the end processing of the `tabular` and `array` environments. A document class can define these hooks as needed.

We proceed with care to make further patches to support tabulars that break over pages. Our patches will not necessarily be effective for other packages that replace the LaTeX `array` and `tabular` environments. I know of none that do so.

```
\appdef\document@inithook{%
 \@ifpackageloaded{array}{\switch@array}{\switch@tabular}%
 \prepdef\endtabular{\endtabular@hook}%
 \@provide\endtabular@hook{}%
 \prepdef\endarray{\endarray@hook}%
 \@provide\endarray@hook{}%
 \providecommand\array@hook{}%
```

Install, effectively, a head patch to `\tabular`. In order to avoid interference from, e.g., the `array` package, we must perform this patch only *after* packages load.

```
 \prepdef\@tabular{\tabular@hook}%
 \@provide\tabular@hook{}%
}%
```

`\switch@tabular` `\switch@array` The two procedures `\switch@tabular` and `\switch@array` apply needed patches to the various tabular procedures, the former applying to the LaTeX kernel, the latter to the required `array` package (and to the number of other required packages that load it).

```
\def\switch@tabular{%
 \let\@array@sw\@array@sw@array
 \@ifx{\@array\@array@LaTeX}{%
  \@ifx{\multicolumn\multicolumn@LaTeX}{%
   \@ifx{\@tabular\@tabular@LaTeX}{%
    \@ifx{\@tabarray\@tabarray@LaTeX}{%
     \@ifx{\array\array@LaTeX}{%
      \@ifx{\endarray\endarray@LaTeX}{%
       \@ifx{\endtabular\endtabular@LaTeX}{%
        \@ifx{\@mkpream\@mkpream@LaTeX}{%
         \@ifx{\@addamp\@addamp@LaTeX}{%
```

```
\@ifx{\@arrayacol\@arrayacol@LaTeX}{%
 \@ifx{\@tabacol\@tabacol@LaTeX}{%
  \@ifx{\@arrayclassz\@arrayclassz@LaTeX}{%
   \@ifx{\@tabclassiv\@tabclassiv@LaTeX}{%
    \@ifx{\@arrayclassiv\@arrayclassiv@LaTeX}{%
     \@ifx{\@tabclassz\@tabclassz@LaTeX}{%
      \@ifx{\@classv\@classv@LaTeX}{%
       \@ifx{\hline\hline@LaTeX}{%
        \@ifx{\@tabularcr\@tabularcr@LaTeX}{%
         \@ifx{\@xtabularcr\@xtabularcr@LaTeX}{%
          \@ifx{\@xargarraycr\@xargarraycr@LaTeX}{%
           \@ifx{\@yargarraycr\@yargarraycr@LaTeX}{%
            \true@sw
           }{%
            \false@sw
           }%
          }{%
           \false@sw
          }%
         }{%
          \false@sw
         }%
        }{%
         \false@sw
        }%
       }{%
        \false@sw
       }%
      }{%
       \false@sw
      }%
     }{%
      \false@sw
     }%
    }{%
     \false@sw
    }%
   }{%
    \false@sw
   }%
  }{%
   \false@sw
  }%
 }{%
  \false@sw
 }%
}{%
```

```
          \false@sw
         }%
        }{%
         \false@sw
        }%
       }{%
        \false@sw
       }%
      }{%
       \false@sw
      }%
     }{%
      \false@sw
     }%
    }{%
     \false@sw
    }%
   }{%
    \false@sw
   }%
  }{%
   \false@sw
  }%
 }{%
  \class@info{Patching LaTeX tabular.}%
 }{%
  \class@info{Unrecognized LaTeX tabular. Please update this document class! (Proceeding with f
 }%
 \let\@array\@array@ltx
 \let\multicolumn\multicolumn@ltx
 \let\@tabular\@tabular@ltx
 \let\@tabarray\@tabarray@ltx
 \let\array\array@ltx
 \let\endarray\endarray@ltx
 \let\endtabular\endtabular@ltx
 \let\@mkpream\@mkpream@ltx
 \let\@addamp\@addamp@ltx
 \let\@arrayacol\@arrayacol@ltx
 \let\@tabacol\@tabacol@ltx
 \let\@arrayclassz\@arrayclassz@ltx
 \let\@tabclassiv\@tabclassiv@ltx
 \let\@arrayclassiv\@arrayclassiv@ltx
 \let\@tabclassz\@tabclassz@ltx
 \let\@classv\@classv@ltx
 \let\hline\hline@ltx
 \let\@tabularcr\@tabularcr@ltx
 \let\@xtabularcr\@xtabularcr@ltx
```

```
 \let\@xargarraycr\@xargarraycr@ltx
 \let\@yargarraycr\@yargarraycr@ltx
}%

\def\switch@array{%
 \@ifpackageloaded{colortbl}{\let\switch@array@info\colortbl@message}{\let\switch@array@info\ar
 \let\@array@sw\@array@sw@LaTeX
 \@ifx{\@array\@array@array}{%
  \@ifx{\@tabular\@tabular@array}{%
   \@ifx{\@tabarray\@tabarray@array}{%
    \@ifx{\array\array@array}{%
     \@ifx{\endarray\endarray@array}{%
      \@ifx{\endtabular\endtabular@array}{%
       \@ifx{\@mkpream\@mkpream@array}{%
        \@ifx{\@classx\@classx@array}{%
         \@ifx{\insert@column\insert@column@array}{%
          \@ifx{\@arraycr\@arraycr@array}{%
           \@ifx{\@xarraycr\@xarraycr@array}{%
            \@ifx{\@xargarraycr\@xargarraycr@array}{%
             \@ifx{\@yargarraycr\@yargarraycr@array}{%
              \true@sw
             }{%
              \false@sw
             }%
            }{%
             \false@sw
            }%
           }{%
            \false@sw
           }%
          }{%
           \false@sw
          }%
         }{%
          \false@sw
         }%
        }{%
         \false@sw
        }%
       }{%
        \false@sw
       }%
      }{%
       \false@sw
      }%
     }{%
      \false@sw
     }%
    }{%
     \false@sw
```

```
    }%
   }{%
    \false@sw
   }%
  }{%
   \false@sw
  }%
 }{%
  \false@sw
 }{%
  \class@info{Patching array package.}%
 }{%
  \switch@array@info
 }%
 \let\@array    \@array@array@new
 \let\@@array   \@array % Cosi fan tutti
 \let\@tabular  \@tabular@array@new
 \let\@tabarray \@tabarray@array@new
 \let\array     \array@array@new
 \let\endarray  \endarray@array@new
 \let\endtabular\endtabular@array@new
 \let\@mkpream  \@mkpream@array@new
 \let\@classx   \@classx@array@new
 \let\@arrayacol\@arrayacol@ltx
 \let\@tabacol  \@tabacol@ltx
 \let\insert@column\insert@column@array@new
 \expandafter\let\csname endtabular*\endcsname\endtabular % Cosi fan tutti
 \let\@arraycr  \@arraycr@new
 \let\@xarraycr \@xarraycr@new
 \let\@xargarraycr\@xargarraycr@new
 \let\@yargarraycr\@yargarraycr@new
}%
\def\array@message{%
 \class@info{Unrecognized array package. Please update this document class! (Proceeding with fi
}%
\def\colortbl@message{%
 \class@info{colortbl package is loaded. (Proceeding with fingers crossed.)}%
}%
```

`\@array@sw` The Boolean `\@array@sw` must be different depending on whether the `array` package is loaded.

```
\def\@array@sw@LaTeX{\@ifx{\\\@tabularcr}}%
\def\@array@sw@array{\@ifx{\d@llarbegin\begingroup}}%
```

`\@tabular` We provide the old versions of `\@tabular` along with the respective new versions. The change here is to avoid committing to LR mode. That will be done later (as late as possible, naturally).

Compatibility note: I had done `\let \col@sep \@undefined` here, but this was not compatible with `colortbl`. I have removed that statement.

```
\def\@tabular@LaTeX{%
 \leavevmode
 \hbox\bgroup$%
  \let\@acol\@tabacol
  \let\@classz\@tabclassz
  \let\@classiv\@tabclassiv
  \let\\\@tabularcr
  \@tabarray
}%
\def\@tabular@ltx{%
  \let\@acoll\@tabacoll
  \let\@acolr\@tabacolr
  \let\@acol\@tabacol
  \let\@classz\@tabclassz
  \let\@classiv\@tabclassiv
  \let\\\@tabularcr
  \@tabarray
}%
\def\@tabular@array{%
 \leavevmode
 \hbox\bgroup$%
  \col@sep\tabcolsep
  \let\d@llarbegin\begingroup
  \let\d@llarend\endgroup
  \@tabarray
}%
\def\@tabular@array@new{%
  \let\@acoll\@tabacoll
  \let\@acolr\@tabacolr
  \let\@acol\@tabacol
```

sepundefined

```
  \let\d@llarbegin\begingroup
  \let\d@llarend\endgroup
  \@tabarray
}%
```

`\@tabarray` Here we provide old and new versions of the `\@tabarray` procedure. The change here is to parametrize the default vertical alignment, which is 'c' in standard LaTeX. Under some circumstances, we want to change this to, say, 'v'.

FIXME: must decouple `array` and `tabular`. Done (it seems).

Note on `colortbl`: this package head-patches `\@tabarray`with its own command `\CT@start`, and tails onto `\endarray` with `\CT@end`. It fortuitously does the former at `\AtBeginDocument` time, and, fortuitously, we do not patch `\endarray`, which it overwrites.

```
\def\@tabarray@LaTeX{%
 \m@th\@ifnextchar[\@array{\@array[c]}%
}%
\def\@tabarray@ltx{%
```

```
 \m@th\@ifnextchar[\@array{\expandafter\@array\expandafter[\array@default]}%
}%
\def\@tabarray@array{%
 \@ifnextchar[{\@@array}{\@@array[c]}%
}%
\def\@tabarray@array@new{%
 \@ifnextchar[{\@@array}{\expandafter\@@array\expandafter[\array@default]}%
}%
```

\@tabularcr
\@tbpen
\@tabularcr
\@xtabularcr
\@xargarraycr
\@yargarraycr
\@arraycr
\@xarraycr

We provide for the \\ command within `tabular` to provide control over page breaking, just the same as that of `eqnarray`.

The count register `\intertabularlinepenalty` is similar to `\interdisplaylinepenalty`: it is the penalty associated with each row of a tabular. When it is set to `\@M`, the tabular will cleave together.

The count register `\@tbpen` is similar to `\@eqpen`: it memorizes the penalty to use after the current tabular row. If the \\ command is in its star form, then `\@eqpen` is set to `\@M`.

We append code to `\samepage` so that a tabular within its scope will cleave together.

We keep the standard definition of `\@tabularcr` in `\@tabularcr@LaTeX` for reference, and provide a new definition that works like `\@eqncr`: it sets `\@tbpen` to `\@M` if the star was given.

We also provide new versions of `\@xtabularcr`, `\@xargarraycr`, and `\@yargarraycr`, all of which invoke `\@tbpen`.

The `\switch@tabular` procedure switches in the new definitions.

```
\newcount\intertabularlinepenalty
\intertabularlinepenalty=100
\newcount\@tbpen
\appdef\samepage{\intertabularlinepenalty\@M}%
\def\@tabularcr@LaTeX{{\ifnum 0=‘}\fi \@ifstar \@xtabularcr \@xtabularcr}%
\def\@tabularcr@ltx{{\ifnum 0=‘}\fi \@ifstar {\global \@tbpen \@M \@xtabularcr }{\global \@tbpe
\def\@xtabularcr@LaTeX{\@ifnextchar [\@argtabularcr {\ifnum 0=‘{\fi }\cr }}%
\def\@xtabularcr@ltx{\@ifnextchar [\@argtabularcr {\ifnum 0=‘{\fi }\cr \noalign {\penalty \@tbp
\def\@xargarraycr@LaTeX#1{\@tempdima #1\advance \@tempdima \dp \@arstrutbox \vrule \@height \z@
\def\@xargarraycr@ltx#1{\@tempdima #1\advance \@tempdima \dp \@arstrutbox \vrule \@height \z@ \
\def\@yargarraycr@LaTeX#1{\cr \noalign {\vskip #1}}%
\def\@yargarraycr@ltx#1{\cr \noalign {\penalty \@tbpen \vskip #1}}%
```

If the `array` package has been loaded, we must alter the meanings of `\@arraycr`, `\@xarraycr`, `\@xargarraycr`, and `\@yargarraycr`. In this case, it is `\switch@array` that switches in the new definitions.

```
\def\@arraycr@array{%
 \relax
 \iffalse{\fi\ifnum 0=‘}\fi
 \@ifstar \@xarraycr \@xarraycr
}%
\def\@arraycr@new{%
 \relax
```

```
 \iffalse{\fi\ifnum 0=`}\fi
 \@ifstar {\global \@tbpen \@M \@xarraycr }{\global \@tbpen \intertabularlinepenalty \@xarraycr
}%
\def\@xarraycr@array{%
 \@ifnextchar [%]
 \@argarraycr {\ifnum 0=`{}\fi\cr}%
}%
\def\@xarraycr@new{%
 \@ifnextchar [%]
 \@argarraycr {\ifnum 0=`{}\fi\cr \noalign {\penalty \@tbpen }}%
}%
\def\@xargarraycr@array#1{%
 \unskip
 \@tempdima #1\advance\@tempdima \dp\@arstrutbox
 \vrule \@depth\@tempdima \@width\z@
 \cr
}%
\def\@xargarraycr@new#1{%
 \unskip
 \@tempdima #1\advance\@tempdima \dp\@arstrutbox
 \vrule \@depth\@tempdima \@width\z@
 \cr
 \noalign {\penalty \@tbpen }%
}%
\def\@yargarraycr@array#1{%
 \cr
 \noalign{\vskip #1}%
}%
\def\@yargarraycr@new#1{%
 \cr
 \noalign{\penalty \@tbpen \vskip #1}%
}%
```

`\array` We provide old and new versions of the `\array` procedure for both LaTeX and the `array` package. The change here is to accomodate the new procedures that will be called for the array boundaries, even though at present they are not special. A thought: here is where matrices can be readily accomodated.

```
\def\array@LaTeX{%
 \let\@acol\@arrayacol
 \let\@classz\@arrayclassz
 \let\@classiv\@arrayclassiv
 \let\\\@arraycr
 \let\@halignto\@empty
 \@tabarray
}%
\def\array@ltx{%
 \@ifmmode{}{\@badmath$}%
 \let\@acoll\@arrayacol
 \let\@acolr\@arrayacol
 \let\@acol\@arrayacol
```

```
 \let\@classz\@arrayclassz
 \let\@classiv\@arrayclassiv
 \let\\\@arraycr
 \let\@halignto\@empty
 \@tabarray
}%
\def\array@array{%
 \col@sep\arraycolsep
 \def\d@llarbegin{$}\let\d@llarend\d@llarbegin\gdef\@halignto{}%
 \@tabarray
}
\def\array@array@new{%
 \@ifmmode{}{\@badmath$}%
 \let\@acoll\@arrayacol
 \let\@acolr\@arrayacol
 \let\@acol\@arrayacol
```

Removed: `\let\col@sep\@undefined`

```
 \def\d@llarbegin{$}%
 \let\d@llarend\d@llarbegin
 \gdef\@halignto{}%
 \@tabarray
}%
```

`\@array` Here we provide old and new versions of `\@array`. The change here is to provide a convenient, flexible, and extensible mechanism for new vertical alignment options.

Instead of testing the optional argument with `\if`, we use a dispatcher based on `\csname`.

We also refrain from using `\ialign`, which would set the `\tabskip` to the wrong value.

Finally, the procedure to set the `\@arstrutbox` is broken out so that it can be patched.

```
\def\@array@LaTeX[#1]#2{%
  \if #1t\vtop \else \if#1b\vbox \else \vcenter \fi\fi
  \bgroup
  \setbox\@arstrutbox\hbox{%
    \vrule \@height\arraystretch\ht\strutbox
           \@depth\arraystretch \dp\strutbox
           \@width\z@}%
  \@mkpream{#2}%
  \edef\@preamble{%
    \ialign \noexpand\@halignto
      \bgroup \@arstrut \@preamble \tabskip\z@skip \cr}%
  \let\@startpbox\@@startpbox \let\@endpbox\@@endpbox
  \let\tabularnewline\\%
    \let\par\@empty
    \let\@sharp##%
    \set@typeset@protect
    \lineskip\z@skip\baselineskip\z@skip
```

```
    \ifhmode \@preamerr\z@ \@@par\fi
    \@preamble
}%
\def\@array@ltx[#1]#2{%
 \@nameuse{@array@align@#1}%
  \set@arstrutbox
  \@mkpream{#2}%
  \prepdef\@preamble{%
    \tabskip\tabmid@skip
    \@arstrut
  }%
  \appdef\@preamble{%
    \tabskip\tabright@skip
    \cr
    \array@row@pre
  }%
% \let\@startpbox\@@startpbox
% \let\@endpbox\@@endpbox
  \let\tabularnewline\\%
  \let\par\@empty
  \let\@sharp##%
  \set@typeset@protect
  \lineskip\z@skip\baselineskip\z@skip
  \tabskip\tableft@skip\relax
  \ifhmode \@preamerr\z@ \@@par\fi
  \everycr{}%
  \expandafter\halign\expandafter\@halignto\expandafter\bgroup\@preamble
}%
%
\def\set@arstrutbox{%
  \setbox\@arstrutbox\hbox{%
    \vrule \@height\arraystretch\ht\strutbox
           \@depth\arraystretch \dp\strutbox
           \@width\z@
  }%
}%
```

\@array@array

```
\def\@array@array[#1]#2{%
  \@tempdima \ht \strutbox
  \advance \@tempdima by\extrarowheight
  \setbox \@arstrutbox \hbox{\vrule
             \@height \arraystretch \@tempdima
             \@depth \arraystretch \dp \strutbox
             \@width \z@}%
  \begingroup
  \@mkpream{#2}%
  \xdef\@preamble{\noexpand \ialign \@halignto
                  \bgroup \@arstrut \@preamble
                          \tabskip \z@ \cr}%
```

```
  \endgroup
  \@arrayleft
  \if #1t\vtop \else \if#1b\vbox \else \vcenter \fi \fi
  \bgroup
  \let \@sharp ##\let \protect \relax
  \lineskip \z@
  \baselineskip \z@
  \m@th
  \let\\\@arraycr \let\tabularnewline\\\let\par\@empty \@preamble
}%
\def\@array@array@new[#1]#2{%
  \@tempdima\ht\strutbox
  \advance\@tempdima by\extrarowheight
  \setbox\@arstrutbox\hbox{%
   \vrule \@height\arraystretch\@tempdima
          \@depth \arraystretch\dp\strutbox
          \@width \z@
  }%
  \begingroup
   \@mkpream{#2}%
   \xdef\@preamble{\@preamble}%
  \endgroup
  \prepdef\@preamble{%
   \tabskip\tabmid@skip
    \@arstrut
  }%
  \appdef\@preamble{%
   \tabskip\tabright@skip
   \cr
   \array@row@pre
  }%
  \@arrayleft
  \@nameuse{@array@align@#1}%
  \m@th
  \let\\\@arraycr
  \let\tabularnewline\\%
  \let\par\@empty
  \let\@sharp##%
  \set@typeset@protect
  \lineskip\z@\baselineskip\z@
  \tabskip\tableft@skip
  \everycr{}%
  \expandafter\halign\expandafter\@halignto\expandafter\bgroup\@preamble
}%
```

`\endarray` Here we provide old and new versions of `\endarray`. The change here is to use a single procedure to close out any array-like structure, namely `\endarray@ltx`. It merely closes out the `\halign`.

```
\def\endarray@LaTeX{%
 \crcr\egroup\egroup
```

```
}%
\def\endarray@ltx{%
 \crcr\array@row@pst\egroup\egroup
}%
\def\endarray@array{%
 \crcr \egroup \egroup \@arrayright \gdef\@preamble{}%
}%
\def\endarray@array@new{%
 \crcr\array@row@pst\egroup\egroup % Same as \endarray@ltx
 \@arrayright
 \global\let\@preamble\@empty
}%
```

`\endtabular`

```
\def\endtabular@LaTeX{%
 \crcr\egroup\egroup $\egroup
}%
\def\endtabular@ltx{%
 \endarray
}%
\def\endtabular@array{%
 \endarray $\egroup
}%
\def\endtabular@array@new{%
 \endarray
}%
```

`endtabular*` Here we provide a proper definition for the star-form of `\end{endtabular}`. It is one of the enduring curiosities that the LaTeX kernel continues to use dangerously and inappropriately "optimized" definitions for such commands.

```
\@namedef{endtabular*}{\endtabular}%
```

`\multicolumn`

```
\long\def\multicolumn@LaTeX#1#2#3{%
 \multispan{#1}\begingroup
  \@mkpream{#2}%
  \def\@sharp{#3}\set@typeset@protect
  \let\@startpbox\@@startpbox\let\@endpbox\@@endpbox
  \@arstrut \@preamble\hbox{}\endgroup\ignorespaces
}%
\long\def\multicolumn@ltx#1#2#3{%
 \multispan{#1}%
 \begingroup
  \@mkpream{#2}%
  \def\@sharp{#3}%
  \set@typeset@protect
 %\let\@startpbox\@@startpbox\let\@endpbox\@@endpbox
  \@arstrut
  \@preamble
```

```
  \hbox{}%
 \endgroup
 \ignorespaces
}%
```

\@array@align@
\array@default

Here are the various procedures for the vertical alignment options. The change from standard LaTeX is that we do not go into math mode in every case: only when required by \vcenter. Also, we use \aftergroup to close out the boxes and modes we have started. It requires only that each procedure issue exactly one unmatched \bgroup.

We establish here the default vertical alignment.

```
\def\@array@align@t{\leavevmode\vtop\bgroup}%
\def\@array@align@b{\leavevmode\vbox\bgroup}%
\def\@array@align@c{\leavevmode\@ifmmode{\vcenter\bgroup}{$\vcenter\bgroup\aftergroup$\aftergro
\def\@array@align@v{%
 \@ifmmode{%
  \@badmath
  \vcenter\bgroup
 }{%
  \@ifinner{%
   $\vcenter\bgroup\aftergroup$
  }{%
   \@@par\bgroup
  }%
 }%
}%
\def\array@default{c}%
```

\array@row@pre
\array@row@pst
\array@row@rst

The procedure \array@row@rst reestablishes a default context for an alignment, so that they can be nested. Any environment or procedure that alters the way alignments are formatted must patch this procedure to restore from that alteration. To start things off, we equate \@array@align@v to \@array@align@c, because it does not make sense to do the former in any context other than the MVL or in a list that will be unboxed onto the MVL.

```
\def\array@row@rst{%
 \let\@array@align@v\@array@align@c
}%
\def\array@row@pre{}%
\def\array@row@pst{}%
```

\toprule
\colrule
\botrule

Default definitions for \toprule, \colrule, \botrule

```
\newcommand\toprule{\tab@rule{\column@font}{\column@fil}{\frstrut}}%
\newcommand\colrule{\unskip\lrstrut\\\tab@rule{\body@font}{}{\frstrut}}%
\newcommand\botrule{\unskip\lrstrut\\\noalign{\hline@rule}{}}%
```

\hline

```
\def\hline@LaTeX{%
 \noalign{\ifnum0=`}\fi\hrule \@height \arrayrulewidth \futurelet
```

```
    \reserved@a\@xhline
}%
\def\hline@ltx{%
 \noalign{%
  \ifnum0=`}\fi
  \hline@rule
  \futurelet\reserved@a\@xhline
 % \noalign ended in \@xhline
}%
\def\@xhline@unneeded{%
 \say\reserved@a
 \ifx\reserved@a\hline
  \vskip\doublerulesep
  \vskip-\arrayrulewidth
 \fi
 \ifnum0=`{\fi}%
}%
\def\tab@rule#1#2#3{%
 \crcr
 \noalign{%
  \hline@rule
  \gdef\@arstrut@hook{%
   \global\let\@arstrut@hook\@empty
   #3%
  }%
  \gdef\cell@font{#1}%
  \gdef\cell@fil{#2}%
 }%
}%
\def\column@font{}%
\def\column@fil{}%
\def\body@font{}%
\def\cell@font{}%
\def\frstrut{}%
\def\lrstrut{}%
```

`\@arstrut@hline` `\@arstrut@org` `\@arstrut@hook` `\@arstrutbox@hline` `\set@arstrutbox` `\hline@rule`

The procedure `\@arstrut@hline` is substantially the same as `\@arstrut`, except the strut copied in is `\@arstrutbox@hline`instead of `\@arstrutbox`.

The procedure `\@arstrut@hook` is redefined in `\tab@rule`!

The register `\@arstrutbox@hline`.

We append to `\set@arstrutbox` the code necessary to set a strut following an `\hline`.

The procedure `\hline@rule` lays down a rule, and changes the meaning of `\@arstrut` so that the next line will be correctly strutted.

The `\@arstrut@hline@clnc` is a klootch, a magic number.

```
\def\@arstrut@hline{%
 \relax
 \@ifmmode{\copy}{\unhcopy}\@arstrutbox@hline
 \@arstrut@hook
```

```
}%
%
\let\@arstrut@org\@arstrut
\def\@arstrut@hook{%
 \global\let\@arstrut\@arstrut@org
}%
%
\newbox\@arstrutbox@hline
\appdef\set@arstrutbox{%
  \setbox\@arstrutbox@hline\hbox{%
    \setbox\z@\hbox{$0^{0}_{}$}%
    \dimen@\ht\z@\advance\dimen@\@arstrut@hline@clnc
    \@ifdim{\dimen@<\arraystretch\ht\strutbox}{\dimen@=\arraystretch\ht\strutbox}{}%
    \vrule \@height\dimen@
           \@depth\arraystretch \dp\strutbox
           \@width\z@
  }%
}%
%
\def\hline@rule{%
 \hrule \@height \arrayrulewidth
 \global\let\@arstrut\@arstrut@hline
}%
\def\@arstrut@hline@clnc{2\p@}% % Klootch: magic number
```

`\tableft@skip`

```
\def\tableft@skip{\z@skip}%
\def\tabmid@skip{\z@skip}%\@flushglue
\def\tabright@skip{\z@skip}%
\def\tableftsep{\tabcolsep}%
\def\tabmidsep{\tabcolsep}%
\def\tabrightsep{\tabcolsep}%
\def\cell@fil{}%
\def\pbox@hook{}%
```

`\@arstrut`

```
\appdef\@arstrut{\@arstrut@hook}%
\let\@arstrut@hook\@empty
\def\@addtopreamble{\appdef\@preamble}%
```

`\@mkpream`

```
\def\@mkpream@LaTeX#1{%
  \@firstamptrue\@lastchclass6
  \let\@preamble\@empty
  \let\protect\@unexpandable@protect
  \let\@sharp\relax
  \let\@startpbox\relax\let\@endpbox\relax
  \@expast{#1}%
  \expandafter\@tfor \expandafter
```

```
    \@nextchar \expandafter:\expandafter=\reserved@a\do
       {\@testpach\@nextchar
    \ifcase \@chclass \@classz \or \@classi \or \@classii \or \@classiii
      \or \@classiv \or\@classv \fi\@lastchclass\@chclass}%
  \ifcase \@lastchclass \@acol
      \or \or \@preamerr \@ne\or \@preamerr \tw@\or \or \@acol \fi
}%
\def\@mkpream@ltx#1{%
 \@firstamptrue
 \@lastchclass6
 \let\@preamble\@empty
 \let\protect\@unexpandable@protect
 \let\@sharp\relax
%\let\@startpbox\relax\let\@endpbox\relax
 \@expast{#1}%
 \expandafter\@tfor\expandafter\@nextchar\expandafter:\expandafter=\reserved@a
 \do{%
  \expandafter\@testpach\expandafter{\@nextchar}%
  \ifcase\@chclass
   \@classz
  \or
   \@classi
  \or
   \@classii
  \or
   \@classiii
  \or
   \@classiv
  \or
   \@classv
  \fi
  \@lastchclass\@chclass
 }%
 \ifcase\@lastchclass
  \@acolr % right-hand column
 \or
 \or
  \@preamerr\@ne
 \or
  \@preamerr\tw@
 \or
 \or
  \@acolr % right-hand column
 \fi
}%
```

\insert@column

```
\def\insert@column@array{%
   \the@toks \the \@tempcnta
   \ignorespaces \@sharp \unskip
```

```
    \the@toks \the \count@ \relax
}%
\def\insert@column@array@new{%
 \the@toks\the\@tempcnta
 \array@row@rst\cell@font
 \ignorespaces\@sharp\unskip
 \the@toks\the\count@
 \relax
}%
```

`\@mkpream@relax` The procedure `\@mkpream@relax` participates in a strange and wonderful method of binding the alignment procedure—but only certain parts thereof.

Here is how it works: in LaTeX, the `array` package, and in the `longtable` package alike, there is a need to create an alignment preamble (using `\@mkpream`) for use by the upcoming `\halign`. Then, in both `array` and `longtable`, TeX's `\edef` is used to 'compile in place' that alignment preamble.

In the case of `array`, the operation is done in order to pre-expand the use of `*`; in `longtable`, it is to set the widths of the columns.

Now, during this `\edef`, certain control sequence names must *not* be expanded, and those are robustified by `\@mkpream@relax`.

```
\def\@mkpream@relax{%
 \let\tableftsep   \relax
 \let\tabmidsep    \relax
 \let\tabrightsep  \relax
 \let\array@row@rst\relax
 \let\cell@font    \relax
 \let\@startpbox   \relax
}%
```

`\@mkpream` We insert `\@mkpream@relax` at the head of the procedure. The robustifying of `\@startpbox` and `\@endpbox` is taken over by this mechanism. We also invoke `\@acolr` instead of `\@acol` when a right-hand column is at hand.

Note on `colortbl`: this package head-patches `\@mkpream` to robustify a number of its commands during the construction of the alignment preamble. The best we can do is to supplement the `\@mkpream@relax` procedure to perform this action.

```
\def\@mkpream@array#1{%
   \gdef\@preamble{}\@lastchclass 4 \@firstamptrue
   \let\@sharp\relax \let\@startpbox\relax \let\@endpbox\relax
   \@temptokena{#1}\@tempswatrue
   \@whilesw\if@tempswa\fi{\@tempswafalse\the\NC@list}%
   \count@\m@ne
   \let\the@toks\relax
   \prepnext@tok
   \expandafter \@tfor \expandafter \@nextchar
    \expandafter :\expandafter =\the\@temptokena \do
   {\@testpach
   \ifcase \@chclass \@classz \or \@classi \or \@classii
     \or \save@decl \or \or \@classv \or \@classvi
```

```
    \or \@classvii \or \@classviii
    \or \@classx
    \or \@classx \fi
  \@lastchclass\@chclass}%
  \ifcase\@lastchclass
  \@acol \or
  \or
  \@acol \or
  \@preamerr \thr@@ \or
  \@preamerr \tw@ \@addtopreamble\@sharp \or
  \or
  \else  \@preamerr \@ne \fi
  \def\the@toks{\the\toks}%
}%
\def\@mkpream@array@new#1{%
 \gdef\@preamble{}%
 \@lastchclass\f@ur
 \@firstamptrue
 \let\@sharp\relax
 \@mkpream@relax
%\let\@startpbox\relax\let\@endpbox\relax
 \@temptokena{#1}\@tempswatrue
 \@whilesw\if@tempswa\fi{\@tempswafalse\the\NC@list}%
 \count@\m@ne
 \let\the@toks\relax
 \prepnext@tok
 \expandafter\@tfor\expandafter\@nextchar\expandafter:\expandafter=\the\@temptokena
 \do{%
  \@testpach
  \ifcase\@chclass
   \@classz
  \or
   \@classi
  \or
   \@classii
  \or
   \save@decl
  \or
  \or
   \@classv
  \or
   \@classvi
  \or
   \@classvii
  \or
   \@classviii
  \or
   \@classx
  \or
   \@classx
```

```
  \fi
  \@lastchclass\@chclass
 }%
 \ifcase\@lastchclass
  \@acolr % right-hand column
 \or
 \or
  \@acolr % right-hand column
 \or
  \@preamerr\thr@@
 \or
  \@preamerr\tw@\@addtopreamble\@sharp
 \or
 \or
 \else
  \@preamerr\@ne
 \fi
 \def\the@toks{\the\toks}%
}%
```

`\@mkpream@relax` David P. Carlisle's `colortbl` package headpatches `\@mkpream` in place during package loading, so it does not know whom it is working on. Let us try to accomodate this package by doing what it would have liked to have done.

Note: it would be far better to break out this mechanism in the `array` package.

```
\appdef\@mkpream@relax{%
 \let\CT@setup       \relax
 \let\CT@color       \relax
 \let\CT@do@color    \relax
 \let\color          \relax
 \let\CT@column@color\relax
 \let\CT@row@color   \relax
 \let\CT@cell@color  \relax
}%
```

`\@addamp`

```
\def\@addamp@LaTeX{%
  \if@firstamp\@firstampfalse\else\edef\@preamble{\@preamble &}\fi
}%
\def\@addamp@ltx{%
 \if@firstamp\@firstampfalse\else\@addtopreamble{&}\fi
}%
```

`\@arrayacol`

```
\def\@arrayacol@LaTeX{%
 \edef\@preamble{\@preamble \hskip \arraycolsep}%
}%
\def\@arrayacol@ltx{%
 \@addtopreamble{\hskip\arraycolsep}%
}%
```

\@tabacol

```
\def\@tabacoll{%
 \@addtopreamble{\hskip\tableftsep\relax}%
}%
\def\@tabacol@LaTeX{%
 \edef\@preamble{\@preamble \hskip \tabcolsep}%
}%
\def\@tabacol@ltx{%
 \@addtopreamble{\hskip\tabmidsep\relax}%
}%
\def\@tabacolr{%
 \@addtopreamble{\hskip\tabrightsep\relax}%
}%
```

\@arrayclassz

```
\def\@arrayclassz@LaTeX{%
 \ifcase \@lastchclass \@acolampacol \or \@ampacol \or
   \or \or \@addamp \or
   \@acolampacol \or \@firstampfalse \@acol \fi
 \edef\@preamble{\@preamble
  \ifcase \@chnum
     \hfil$\relax\@sharp$\hfil \or $\relax\@sharp$\hfil
    \or \hfil$\relax\@sharp$\fi}%
}%
\def\@arrayclassz@ltx{%
 \ifcase\@lastchclass
  \@acolampacol
 \or
  \@ampacol
 \or
 \or
 \or
  \@addamp
 \or
  \@acolampacol
 \or
  \@firstampfalse\@acoll
 \fi
 \ifcase\@chnum
  \@addtopreamble{%
   \hfil\array@row@rst$\relax\@sharp$\hfil
  }%
 \or
  \@addtopreamble{%
   \array@row@rst$\relax\@sharp$\hfil
  }%
 \or
  \@addtopreamble{%
   \hfil\array@row@rst$\relax\@sharp$%
  }%
```

```
 \fi
}%
```

\@tabclassz

```
\def\@tabclassz@LaTeX{%
  \ifcase\@lastchclass
    \@acolampacol
  \or
    \@ampacol
  \or
  \or
  \or
    \@addamp
  \or
    \@acolampacol
  \or
    \@firstampfalse\@acol
  \fi
  \edef\@preamble{%
    \@preamble{%
      \ifcase\@chnum
        \hfil\ignorespaces\@sharp\unskip\hfil
      \or
        \hskip1sp\ignorespaces\@sharp\unskip\hfil
      \or
        \hfil\hskip1sp\ignorespaces\@sharp\unskip
      \fi}}%
}%
\def\@tabclassz@ltx{%
 \ifcase\@lastchclass
  \@acolampacol
 \or
  \@ampacol
 \or
 \or
 \or
  \@addamp
 \or
  \@acolampacol
 \or
  \@firstampfalse\@acoll
 \fi
 \ifcase\@chnum
  \@addtopreamble{%
   {\hfil\array@row@rst\cell@font\ignorespaces\@sharp\unskip\hfil}%
  }%
 \or
  \@addtopreamble{%
   {\cell@fil\hskip1sp\array@row@rst\cell@font\ignorespaces\@sharp\unskip\hfil}%
  }%
```

```
  \or
   \@addtopreamble{%
    {\hfil\hskip1sp\array@row@rst\cell@font\ignorespaces\@sharp\unskip\cell@fil}%
   }%
  \fi
}%
```

\@tabclassiv

```
\def\@tabclassiv@LaTeX{%
 \@addtopreamble\@nextchar
}%
\def\@tabclassiv@ltx{%
 \expandafter\@addtopreamble\expandafter{\@nextchar}%
}%
```

\@arrayclassiv

```
\def\@arrayclassiv@LaTeX{%
 \@addtopreamble{$\@nextchar$}%
}%
\def\@arrayclassiv@ltx{%
 \expandafter\@addtopreamble\expandafter{\expandafter$\@nextchar$}%
}%
```

\@classv

```
\def\@classv@LaTeX{%
 \@addtopreamble{\@startpbox{\@nextchar}\ignorespaces
 \@sharp\@endpbox}%
}%
\def\@classv@ltx{%
 \expandafter\@addtopreamble
 \expandafter{%
 \expandafter \@startpbox
 \expandafter {\@nextchar}%
 \pbox@hook\array@row@rst\cell@font\ignorespaces\@sharp\@endpbox
 }%
}%
```

\@classx

```
\def\@classx@array{%
  \ifcase \@lastchclass
  \@acolampacol \or
  \@addamp \@acol \or
  \@acolampacol \or
  \or
  \@acol \@firstampfalse \or
  \@addamp
  \fi
}%
\def\@classx@array@new{%
```

```
\ifcase \@lastchclass
 \@acolampacol
\or
 \@addamp \@acol
\or
 \@acolampacol
\or
\or
 \@firstampfalse\@acoll
\or
 \@addamp
\fi
}%
```

6.15 Repair other broken parts of LaTeX

`\@xbitor` Expansion part has extraneous space token. Removed.

```
\def\@xbitor@LaTeX #1{\@tempcntb \count#1
   \ifnum \@tempcnta =\z@
   \else
     \divide\@tempcntb\@tempcnta
     \ifodd\@tempcntb \@testtrue\fi
   \fi}%
\def\@xbitor@ltx#1{%
 \@tempcntb\count#1\relax
 \@ifnum{\@tempcnta=\z@}{}{%
  \divide\@tempcntb\@tempcnta
  \@ifodd\@tempcntb{\@testtrue}{}%
 }%
}%
\@ifx{\@xbitor\@xbitor@LaTeX}{%
  \class@info{Repairing broken LaTeX \string\@xbitor}%
}{%
  \class@info{Unrecognized LaTeX \string\@xbitor. Please update this document class! (Proceedin
}%
\let\@xbitor\@xbitor@ltx
```

6.16 Syntax

`\@gobble@opt@one` The `\@gobble@opt@one` command eats up an optional argument and one required argument.

```
\newcommand*\@gobble@opt@one[2][]{}%
```

6.17 Auto-indented Contents

Facility to automatically determine the proper indentation of the TOC entries.

Note on `hyperref` compatibility: We must respect that `\contentsline` now has a fourth argument. So, instead of trying to override the meaning of

`\contentsline`, we use the aux file to remember max values from one run to the next.

In this respect, this package retains compatibility with `hyperref`.

`\@starttoc` Install hooks at beginning and end of the TOC processing.

```
\def\@starttoc#1{%
  \begingroup
    \toc@pre
    \makeatletter
    \@input{\jobname.#1}%
    \if@filesw
      \expandafter\newwrite\csname tf@#1\endcsname
      \immediate\openout \csname tf@#1\endcsname \jobname.#1\relax
    \fi
    \@nobreakfalse
    \toc@post
  \endgroup
}%
\def\toc@pre{}%
\def\toc@post{}%
```

`\toc@@font` Interface for setting the formatting characteristics of this part of the TOC.

Note: `\toc@@font` is the common font for all auto-sizing toc commands, although this, too, could become a dispatcher.

```
\def\toc@@font{}%
\def\ltxu@dotsep{\z@}%
```

`\l@section` Interface for determining which TOC elements are automatically indented.

All of the `\l@...` commands simply go through the utility procedure `\l@@sections`. The calling convention is to pass the name of self and the name of parent. If you want to exclude any of these from the indentation scheme, simply leave the `\l@...` command undefined.

Note that the parent of "section" is nil, so we have to define a stub.

```
\def\l@section{\l@@sections{}{section}}% Implicit #3#4

\def\tocleft@{\z@}%

\def\l@subsection{\l@@sections{section}{subsection}}% Implicit #3#4

\def\l@subsubsection{\l@@sections{subsection}{subsubsection}}% Implicit #3#4

\def\l@paragraph{\l@@sections{subsubsection}{paragraph}}% Implicit #3#4

\def\l@subparagraph#1#2{\l@@sections{paragraph}{subparagraph}}% Implicit #3#4
```

Glom some `\dimen` registers.

```
\let\tocdim@section       \leftmargini
\let\tocdim@subsection    \leftmarginii
\let\tocdim@subsubsection \leftmarginiii
```

```
\let\tocdim@paragraph     \leftmarginiv
\let\tocdim@appendix      \leftmarginv
\let\tocdim@pagenum       \leftmarginvi
```

\toc@pre@auto
\toc@post@auto

We patch \@starttoc to: 1) before TOC processing, initialize the max registers and set the needed dimensions from the values stored in the auxiliary file, and 2) after TOC processing, store out those max register values into the auxiliary file.

Note that the font is set here: all other TOC entries must override these font settings.

To activate this override of the standard LaTeX processing, the substyle does: \let\toc@pre\toc@pre@auto and \let\toc@post\toc@post@auto.

```
\def\toc@pre@auto{%
  \toc@@font
  \@tempdima\z@
  \toc@setindent\@tempdima{section}%
  \toc@setindent\@tempdima{subsection}%
  \toc@setindent\@tempdima{subsubsection}%
  \toc@setindent\@tempdima{paragraph}%
  \toc@letdimen{appendix}%
  \toc@letdimen{pagenum}%
}%
\def\toc@post@auto{%
  \if@filesw
   \begingroup
    \toc@writedimen{section}%
    \toc@writedimen{subsection}%
    \toc@writedimen{subsubsection}%
    \toc@writedimen{paragraph}%
    \toc@writedimen{appendix}%
    \toc@writedimen{pagenum}%
   \endgroup
  \fi
}%
```

\toc@setindent

```
\def\toc@setindent#1#2{%
 \csname tocdim@#2\endcsname\tocdim@min\relax
 \@ifundefined{tocmax@#2}{\@namedef{tocmax@#2}{\z@}}{}%
 \advance#1\@nameuse{tocmax@#2}\relax
 \expandafter\edef\csname tocleft@#2\endcsname{\the#1}%
}%
```

\toc@letdimen

```
\def\toc@letdimen#1{%
 \csname tocdim@#1\endcsname\tocdim@min\relax
 \@ifundefined{tocmax@#1}{\@namedef{tocmax@#1}{\z@}}{}%
 \expandafter\let\csname tocleft@#1\expandafter\endcsname\csname tocmax@#1\endcsname
}%
```

\toc@writedimen

```
\def\toc@writedimen#1{%
 \immediate\write\@auxout{%
  \gdef\expandafter\string\csname tocmax@#1\endcsname{%
   \expandafter\the\csname tocdim@#1\endcsname
  }%
 }%
}%
```

\l@@sections The procedure for formatting the indented TOC entries. We use control sequence names such as \tocmax@section and \tocleft@section, the former being written to the auxiliary file and the latter only defined for the duration of the TOC processing.

Note that the assignment of \box\@tempboxa by \set@tocdim@pagenum must endure over the invocation of #3: it contains the page number which will be set just before the \par.

The arguments:

#1 superior section

#2 this section

#3 content, including possible \numberline

#4 page number

```
\def\l@@sections#1#2#3#4{%
 \begingroup
  \everypar{}%
  \set@tocdim@pagenum\@tempboxa{#4}%
  \global\@tempdima\csname tocdim@#2\endcsname
  \leftskip\csname tocleft@#2\endcsname\relax
  \dimen@\csname tocleft@#1\endcsname\relax
  \parindent-\leftskip\advance\parindent\dimen@
  \rightskip\tocleft@pagenum plus 1fil\relax
  \skip@\parfillskip\parfillskip\z@
  \let\numberline\numberline@@sections
  \@nameuse{l@f@#2}%
  \ignorespaces#3\unskip\nobreak\hskip\skip@
  \hb@xt@\rightskip{\hfil\unhbox\@tempboxa}\hskip-\rightskip\hskip\z@skip
```

By side effect, set the value of, e.g., \tocdim@section.

Note that the \par must not be executed before the value of \@tempdima is expanded (outside the current group). Otherwise, the lineno.sty package may interfere (it unfortunately does a global assignment of \@tempdima).

```
  \expandafter\par
  \expandafter\aftergroup\csname tocdim@#2%
  \expandafter\endcsname
  \expandafter\endgroup
               \the\@tempdima\relax
}%
```

In the call to \set@tocdim@pagenum, I am now exposing the use of the particular box register.

```
\def\set@tocdim@pagenum#1#2{%
 \setbox#1\hbox{\ignorespaces#2}%
 \@ifdim{\tocdim@pagenum<\wd#1}{\global\tocdim@pagenum\wd#1}{}%
}%
```

\numberline@@sections The utility procedure for all \numberline processing in indented TOC entries. The first argument is self.

We use \@tempdima to pass a value around (via global assignment) because \numberline executes inside a group if the hyperref package is loaded. Would that it were not so!

```
\def\numberline@@sections#1{%
 \leavevmode\hb@xt@-\parindent{%
  \hfil
  \@if@empty{#1}{}{%
   \setbox\z@\hbox{#1.\kern\ltxu@dotsep}%
   \@ifdim{\@tempdima<\wd\z@}{\global\@tempdima\wd\z@}{}%
   \unhbox\z@
  }%
 }%
 \ignorespaces
}%
\def\tocdim@min{\z@}%
```

6.18 Lists

\list Using \parshape to implement lists was always suspect (can you get behind \parshape\@ne?) and we now see that it was a mistake all along. Why? Because \parshape, like \hangindent, achieves its effect via "shifting" the \hboxes in a paragraph instead of using \leftskip and \parindent, which is robust during column balancing.

We introduce the alternative method with a hook into the LaTeX kernel procedure \list, which is the implementation of all lists.

```
\def\list#1#2{%
  \ifnum \@listdepth >5\relax
    \@toodeep
  \else
    \global\advance\@listdepth\@ne
  \fi
  \rightmargin\z@
  \listparindent\z@
  \itemindent\z@
  \csname @list\romannumeral\the\@listdepth\endcsname
  \def\@itemlabel{#1}%
  \let\makelabel\@mklab
  \@nmbrlistfalse
  #2\relax
```

```
  \@trivlist
  \parskip\parsep
  \set@listindent
  \ignorespaces
}%
\def\set@listindent@parshape{%
 \parindent\listparindent
 \advance\@totalleftmargin\leftmargin
 \advance\linewidth-\rightmargin
 \advance\linewidth-\leftmargin
 \parshape\@ne\@totalleftmargin\linewidth
}%
\def\set@listindent@{%
 \parindent\listparindent
 \advance\@totalleftmargin\leftmargin
 \advance\rightskip\rightmargin
 \advance\leftskip\@totalleftmargin
}%
\let\set@listindent\set@listindent@parshape
```

6.19 Hypertext capabilities

\href
\url

We provide support for the \href, \url, and \doi commands. Packages, like hyperref, may override these definitions and provide better semantics.

\URL@prefix
\doi
\doibase

```
\providecommand\href[0]{\begingroup\@sanitize@url\@href}%
\def\@href#1{\@@startlink{#1}\endgroup\@@href}%
\def\@@href#1{#1\@@endlink}%
\providecommand \url  [0]{\begingroup\@sanitize@url \@url }%
\def \@url #1{\endgroup\@href {#1}{\URL@prefix#1}}%
\providecommand \URL@prefix [0]{URL }%
\providecommand\doi[0]{\begingroup\@sanitize@url\@doi}%
\def\@doi#1{\endgroup\@@startlink{\doibase#1}doi:\discretionary {}{}{}#1\@@endlink }%
\providecommand \doibase [0]{http://dx.doi.org/}%
\providecommand \@sanitize@url[0]{\chardef\cat@space\the\catcode`\ \@sanitize\catcode`\ \cat@sp
```

\@@startlink
\@@endlink
\pdfstartlink@attr
\hypertext@enable@ltx

How we define \@@startlink and \@@endlink will depend on whether we are running under PDFLATEX. If so, and if PDF output is requested, then we use its primitives to implement hypertext, breaking out the link attributes in \pdfstartlink@attr and using the hyperref defaults; \pdfstartlink@attr can be redefined by a client package. Otherwise we fall back the HyperTEX standard and leave things to the DVI translator.

A class or package that wishes to employ hypertext capabilities should execute the \hypertext@enable@ltx procedure.

```
\def\@@startlink#1{}%
\def\@@endlink{}%
\@ifxundefined \pdfoutput {\true@sw}{\@ifnum{\z@=\pdfoutput}{\true@sw}{\false@sw}}%
{%
 \def\@@startlink@hypertext#1{\leavevmode\special{html:<a href="#1">}}%
```

```
 \def\@@endlink@hypertext{\special{html:</a>}}%
}{%
 \def\@@startlink@hypertext#1{%
  \leavevmode
  \pdfstartlink\pdfstartlink@attr
   user{/Subtype/Link/A<</Type/Action/S/URI/URI(#1)>>}%
  \relax
 }%
 \def\@@endlink@hypertext{\pdfendlink}%
 \def\pdfstartlink@attr{attr{/Border[0 0 1 ]/H/I/C[0 1 1]}}%
}%
\def\hypertext@enable@ltx{%
 \let\@@startlink\@@startlink@hypertext
 \let\@@endlink\@@endlink@hypertext
}%
```

\href The `\href` command of `hyperref` was extend somewhere between versions 6.75r and 6.80e. We apply a repair to the earlier version (if present) so that it works like the later version.

The issue is the presence of whitespace, either following the `\href` token or following the first argument's closing brace character.

```
\def\href@Hy{\hyper@normalise \href@ }%
\def\href@Hy@ltx{\@ifnextchar\bgroup\Hy@href{\hyper@normalise\href@}}%
\def\Hy@href#{\hyper@normalise\href@}%
\begingroup
  \endlinechar=-1 %
  \catcode`\^^A=14 %
  \catcode`\^^M\active
  \catcode`\%\active
  \catcode`\#\active
  \catcode`\_\active
  \catcode`\$\active
  \catcode`\&\active
  \gdef\hyper@normalise@ltx{^^A
    \begingroup
    \catcode`\^^M\active
    \def^^M{ }^^A
    \catcode`\%\active
    \let%\@percentchar
    \let\%\@percentchar
    \catcode`\#\active
    \def#{\hyper@hash}^^A
    \def\#{\hyper@hash}^^A
    \@makeother\&^^A
    \edef&{\string&}^^A
    \edef\&{\string&}^^A
    \edef\textunderscore{\string_}^^A
    \let\_\textunderscore
    \catcode`\_\active
```

```
    \let_\textunderscore
    \let~\hyper@tilde
    \let\~\hyper@tilde
    \let\textasciitilde\hyper@tilde
    \let\\\@backslashchar
    \edef${\string$}^^A
    \Hy@safe@activestrue
    \hyper@n@rmalise
  }^^A
  \catcode`\#=6 ^^A
  \gdef\Hy@ActiveCarriageReturn@ltx{^^M}^^A
  \gdef\hyper@n@rmalise@ltx#1#2{^^A
    \def\Hy@tempa{#2}^^A
    \ifx\Hy@tempa\Hy@ActiveCarriageReturn
      \Hy@ReturnAfterElseFi{^^A
        \hyper@@normalise{#1}^^A
      }^^A
    \else
      \Hy@ReturnAfterFi{^^A
        \hyper@@normalise{#1}{#2}^^A
      }^^A
    \fi
  }^^A
  \gdef\hyper@@normalise@ltx#1#2{^^A
    \edef\Hy@tempa{^^A
      \endgroup
      \noexpand#1{\Hy@RemovePercentCr#2%^^M\@nil}^^A
    }^^A
    \Hy@tempa
  }^^A
  \gdef\Hy@RemovePercentCr@ltx#1%^^M#2\@nil{^^A
    #1^^A
    \ifx\limits#2\limits
    \else
      \Hy@ReturnAfterFi{^^A
        \Hy@RemovePercentCr #2\@nil
      }^^A
    \fi
  }^^A
\endgroup
\def\switch@hyperref@href{%
 \expandafter\@ifx\expandafter{\csname href \endcsname\href@Hy}{
  \class@info{Repairing hyperref 6.75r \string\href}%
  \let\hyper@normalise\hyper@normalise@ltx
  \let\hyper@@normalise\hyper@@normalise@ltx
  \let\hyper@n@rmalise\hyper@n@rmalise@ltx
  \let\Hy@ActiveCarriageReturn\Hy@ActiveCarriageReturn@ltx
  \let\Hy@RemovePercentCr\Hy@RemovePercentCr@ltx
  \let\href\href@Hy@ltx
 }{}%
```

```
}%
\appdef\document@inithook{\switch@hyperref@href}%
```

`\typeout` We make the `\typeout` procedure of LaTeX be `\long`, because sometimes we are talking about `\par`.

```
\def\typeout@org#1{%
 \begingroup
  \set@display@protect
  \immediate\write\@unused{#1}%
 \endgroup
}%
\long\def\typeout@ltx#1{%
 \begingroup
  \set@display@protect
  \immediate\write\@unused{#1}%
 \endgroup
}%
\@ifx{\typeout\typeout@org}{%
 \let\typeout\typeout@ltx
 \true@sw
}{%
 \rvtx@ifformat@geq{2020-10-01}%
   {\true@sw}{\false@sw}%
}%
 {\class@info{Making \string\typeout\space \string\long}}%
 {}%
```

6.20 End of the `kernel` DOCSTRIP module

Here ends the module.

```
%</kernel>
```

Index

Symbols

\# 2156, 2167, 2169, 2185
\$ 2158
\% 900, 2155, 2164, 2166
\& 2159, 2170, 2172
\(..... 665
\) 666
.aux 40
.cls 19
.dtx 4
.rtx 19
\@ 478, 486
\@@array 1229, 1295, 1298
\@@end 373, 466
\@@endlink 68
\@@endlink 2120, 2125, 2128
\@@endlink@hypertext ... 2133, 2141, 2146
\@@endpbox 1396, 1419, 1525, 1534
\@@href 2119, 2120
\@@par 795, 971, 1011, 1402, 1426, 1552
\@@startlink 68
\@@startlink .. 2119, 2125, 2128
\@@startlink@hypertext .. 2132, 2135, 2145
\@@startpbox . 1396, 1418, 1525, 1534
\@Esphack 888, 890
\@M 47
\@MM 27
\@acol 57
\@acol 1257, 1266, 1283, 1352, 1363, 1379, 1653, 1654, 1732, 1734, 1832, 1878, 1945, 1948, 1956
\@acolampacol 1830, 1832, 1840, 1848, 1868, 1876, 1892, 1900, 1944, 1946, 1954, 1958
\@acoll .. 1264, 1281, 1361, 1377, 1850, 1902, 1961
\@acolr 57
\@acolr .. 1265, 1282, 1362, 1378, 1683, 1691, 1781, 1784
\@addamp . 1064, 1154, 1805, 1831, 1846, 1874, 1898, 1945, 1949, 1956, 1963
\@addamp@LaTeX 1064, 1805
\@addamp@ltx 1154, 1808
\@addtopreamble 1640, 1736, 1788, 1809, 1815, 1818, 1824, 1827, 1853, 1857, 1861, 1905, 1909, 1913, 1919, 1922, 1925, 1928, 1931, 1935
\@afterindentfalse 932
\@afterindenttrue 930
\@ampacol 1830, 1842, 1870, 1894
\@argarraycr 1324, 1328
\@argswap 13
\@argswap 297
\@argswap@val 13
\@argswap@val 297, 574
\@argtabularcr 1306, 1307
\@array 49
\@array .. 1056, 1146, 1171, 1228, 1229, 1289, 1292, 1385
\@array@LaTeX 1056, 1385
\@array@align@ 1541
\@array@align@b 1542
\@array@align@c 53
\@array@align@c 1543, 1558
\@array@align@t 1541
\@array@align@v 53
\@array@align@v 1544, 1558
\@array@array 1171, 1438
\@array@array@new .. 1228, 1460
\@array@ltx 1146, 1405
\@array@sw 45
\@array@sw 1055, 1170, 1252
\@array@sw@LaTeX ... 1170, 1252
\@array@sw@array ... 1055, 1253
\@arrayacol 1065, 1155, 1237, 1352, 1361–1363, 1377–1379, 1811

\@arrayacol@LaTeX . . 1065, 1811
\@arrayacol@ltx 1155, 1237, 1814
\@arrayclassiv 1069, 1159, 1354, 1365, 1924
\@arrayclassiv@LaTeX 1069, 1924
\@arrayclassiv@ltx . 1159, 1927
\@arrayclassz . 1067, 1157, 1353, 1364, 1829
\@arrayclassz@LaTeX 1067, 1829
\@arrayclassz@ltx . . 1157, 1838
\@arraycr 47
\@arraycr 1180, 1241, 1300, 1355, 1366, 1458, 1484
\@arraycr@array 1180, 1312
\@arraycr@new 1241, 1317
\@arrayleft 1451, 1481
\@arrayright 1501, 1505
\@arstrut 54
\@arstrut 1395, 1411, 1448, 1474, 1526, 1535, 1608, 1610, 1627, 1638
\@arstrut@hline 54
\@arstrut@hline 1602
\@arstrut@hline@clnc 54
\@arstrut@hline@clnc 1617, 1629
\@arstrut@hook 54
\@arstrut@hook 1588, 1589, 1602, 1638, 1639
\@arstrut@org 1602
\@arstrutbox 49, 54
\@arstrutbox 1308, 1309, 1332, 1338, 1388, 1432, 1441, 1463
\@arstrutbox@hline 54
\@arstrutbox@hline 1602
\@auxout 670, 679, 2043
\@backslashchar 2180
\@boole@def 13
\@boole@def 307, 319–322, 324–334
\@boolean 13
\@boolean 307
\@booleanfalse 13
\@booleanfalse . . . 317, 803, 848
\@booleantrue 13
\@booleantrue 317, 733, 846
\@bsphack 678, 893
\@caption 30
\@captype 34
\@captype 749
\@centercr 20, 21
\@centercr 497
\@centering . 504, 507, 510, 529, 535, 548
\@chclass 1651, 1652, 1667, 1680, 1725, 1730, 1756, 1778
\@checkend . . . 369, 411, 423, 427
\@chnum . . 1834, 1852, 1882, 1904
\@classi . 1651, 1670, 1725, 1759
\@classii 1651, 1672, 1725, 1761
\@classiii 1651, 1674
\@classiv 1259, 1268, 1354, 1365, 1652, 1676
\@classv . 1071, 1161, 1652, 1678, 1726, 1766, 1930
\@classv@LaTeX 1071, 1930
\@classv@ltx 1161, 1934
\@classvi 1726, 1768
\@classvii 1727, 1770
\@classviii 1727, 1772
\@classx . 1178, 1236, 1728, 1729, 1774, 1776, 1942
\@classx@array 1178, 1942
\@classx@array@new . 1236, 1952
\@classz . 1258, 1267, 1353, 1364, 1651, 1668, 1725, 1757
\@clsextension 19, 20
\@clsextension . . . 460, 477, 485
\@ctrerr 924
\@currbox 34, 35
\@currbox 840, 843, 845, 846, 852, 853, 862
\@currentHref 615, 1037
\@currentlabel 500, 516, 642, 680
\@currentlabelname 39
\@currext 19
\@currext 442, 459, 460, 476, 484
\@currname 19
\@currname . . . 442, 458, 475, 483
\@currnamestack@ltx . 470, 474, 479, 487, 489
\@dblarg 940, 941

\@dblfloat 784, 786, 808
\@dblfloat@LaTeX 784, 808
\@defaultsubs 392
\@depth . . 1308, 1309, 1333, 1339, 1390, 1434, 1443, 1465, 1620
\@dofilelist 385
\@doi 2124, 2125
\@eha 202
\@enddocumenthook 368
\@endpbox 57
\@endpbox 1396, 1419, 1525, 1534, 1646, 1662, 1716, 1747, 1932, 1939
\@eqcnt . . 503, 508, 509, 511, 517, 530, 533, 536
\@eqncr 47
\@eqncr 505, 519
\@eqnsel 507, 529
\@eqnswtrue 501, 517
\@eqpen 47
\@esphack 682
\@expast 1647, 1663
\@extension 18, 19
\@finalstrut 648
\@firstampfalse 1806, 1809, 1832, 1850, 1878, 1902, 1948, 1961
\@firstamptrue 1642, 1657, 1715, 1744
\@firstofone 367
\@float 29, 33, 35
\@float@LaTeX 782, 783, 807
\@floatboxreset 755
\@flushglue 495, 1631
\@font@warning 387
\@footnotemark 26
\@footnotetext 23, 26
\@footnotetext 587, 607, 608, 623
\@gobble@opt@one 63
\@gobble@opt@one 1985
\@halignto 1356, 1367, 1372, 1382, 1394, 1428, 1447, 1492
\@hang@from 968, 999
\@hang@froms 1008, 1027
\@hangfrom 37–39
\@hangfrom 999
\@hangfrom@section 38
\@height . 1308, 1309, 1389, 1433, 1442, 1464, 1566, 1619, 1626
\@href 2118, 2119, 2122
\@ialph 919, 920
\@if@empty 319, 2076
\@if@sw . . 322, 323, 377, 394, 396
\@ifcat 319
\@ifdim . 319, 386, 865, 964, 1004, 1618, 2071, 2078
\@ifempty 13
\@ifeof 319
\@ifhbox 319, 861, 880
\@ifhmode 319
\@ifinner 319, 712, 1549
\@ifl@t@r 303
\@ifl@ter 462
\@ifmmode . 319, 1360, 1376, 1543, 1545, 1604
\@ifnextchar 557, 566, 581, 1289, 1292, 1295, 1298, 1306, 1307, 1323, 1327, 2149
\@ifnotempty 13
\@ifnotrelax 13
\@ifnotrelax . 297, 685, 860, 879
\@ifnum . . 319, 845, 946, 975, 989, 1974, 2130
\@ifodd 319, 1976
\@ifpackageloaded . . . 221, 606, 1045, 1169
\@ifstar . . 939, 1304, 1305, 1315, 1320
\@ifundefined . 33, 955, 968, 973, 984, 987, 1008, 1013, 1018, 1021, 2033, 2039
\@ifvbox 319
\@ifvmode 319, 551
\@ifvoid 319, 709, 720
\@ifx 13
\@ifx 297, 298, 301, 319, 340, 395, 540, 619, 907, 1056–1076, 1171–1183, 1252, 1253, 1979, 2217, 2240
\@ifx@empty 319
\@ifxundefined 13

\@ifxundefined . . . 269, 274, 279, 297, 782, 2130
\@ifxundefined@cs 301, 790, 801, 821
\@iiiparbox 731
\@itemlabel 2095
\@lastchclass 1642, 1652, 1653, 1658, 1680, 1682, 1715, 1730, 1731, 1743, 1778, 1780, 1830, 1839, 1867, 1891, 1943, 1953
\@latex@warning@no@line .. 397
\@listdepth . . . 2086, 2089, 2094
\@mainaux 15
\@mainaux 378
\@makecol 9
\@makefnmark 23, 27
\@makefnmark 575, 623, 769
\@makefntext 27
\@makefntext 646
\@makeother 900, 904, 2170
\@minipagefalse 728
\@mklab 2096
\@mkpream 57, 59
\@mkpream 1063, 1153, 1177, 1235, 1392, 1408, 1446, 1469, 1523, 1531, 1641, 1714
\@mkpream@LaTeX 1063, 1641
\@mkpream@array 1177, 1714
\@mkpream@array@new 1235, 1741
\@mkpream@ltx 1153, 1656
\@mkpream@relax 57
\@mkpream@relax 1706, 1746, 1796
\@mpargs 731
\@mpfn 23, 28
\@mpfn . . . 560, 564, 573, 586, 592, 596, 599, 602, 643, 742
\@mpfootins 31
\@mpfootins . 698, 701, 702, 709, 710, 713, 716, 720, 721
\@mpfootnotetext 26
\@mpfootnotetext . 609, 610, 623
\@mpmakefntext 734
\@multiplelabels 395, 403
\@namedef . 814, 1520, 2033, 2039
\@nameuse . 218, 737, 1406, 1482, 2034, 2060
\@ne 11, 67
\@newl@bel 380
\@nextchar 1649, 1650, 1664, 1666, 1722, 1753, 1919, 1922, 1925, 1928, 1931, 1938
\@nmbrlistfalse 2097
\@nobreakfalse 1995
\@normalcr 20
\@normalcr 491
\@nul 905, 916
\@onefilewithoptions 18
\@outputdblcol 9
\@p@pfilename 473
\@p@pfilename@ltx 470, 482
\@parboxrestore 641, 754
\@pass@ptions 448
\@percentchar 2165, 2166
\@preamble 1393, 1395, 1403, 1409, 1413, 1428, 1447, 1448, 1458, 1470, 1472, 1476, 1492, 1501, 1506, 1526, 1536, 1640, 1643, 1659, 1715, 1742, 1806, 1812, 1821, 1833, 1880, 1881
\@preamerr 1402, 1426, 1654, 1686, 1688, 1735, 1736, 1738, 1786, 1788, 1792
\@provide . 339, 1047, 1049, 1052
\@pushfilename 473
\@pushfilename@ltx . . . 453, 473
\@refundefined 393
\@reset@ptions 19
\@reset@ptions 456, 471
\@resetactivechars 261
\@rightskip 494
\@runin@to 38
\@runin@to 984, 1000
\@runin@tos 1018, 1028
\@sanitize 2127
\@sanitize@url 2118, 2121, 2124, 2127
\@seccntformat 37, 38
\@seccntformat 956
\@secpenalty 937
\@sect 37

\@sect 944
\@sect@ltx 941, 945
\@setckpt 379
\@sharp . . 1399, 1422, 1454, 1487, 1524, 1532, 1645, 1661, 1696, 1702, 1716, 1736, 1745, 1788, 1835, 1836, 1854, 1858, 1862, 1883, 1885, 1887, 1906, 1910, 1914, 1932, 1939
\@ssect 37, 39
\@ssect 1001
\@ssect@ltx 940, 1001
\@startcolumn 9
\@startpbox 57
\@startpbox 1396, 1418, 1525, 1534, 1646, 1662, 1712, 1716, 1747, 1931, 1937
\@startsection 37, 39
\@startsection 925
\@startsection@hook . . 926, 943
\@starttoc 65
\@starttoc 1986
\@svsec 38
\@svsec . . 948, 954, 956, 959, 969, 985
\@svsechd 982, 1016
\@tabacol 1066, 1156, 1238, 1257, 1266, 1283, 1817
\@tabacol@LaTeX 1066, 1820
\@tabacol@ltx . . 1156, 1238, 1823
\@tabacoll 1264, 1281, 1817
\@tabacolr 1265, 1282, 1826
\@tabarray 46
\@tabarray 1059, 1149, 1173, 1231, 1261, 1270, 1278, 1286, 1288, 1357, 1368, 1373, 1383
\@tabarray@LaTeX . . . 1059, 1288
\@tabarray@array . . . 1173, 1294
\@tabarray@array@new 1231, 1297
\@tabarray@ltx 1149, 1291
\@tabclassiv . 1068, 1158, 1259, 1268, 1918
\@tabclassiv@LaTeX . 1068, 1918
\@tabclassiv@ltx . . . 1158, 1921
\@tabclassz . . 1070, 1160, 1258, 1267, 1866
\@tabclassz@LaTeX . . 1070, 1866
\@tabclassz@ltx 1160, 1890
\@tabular 45
\@tabular 1051, 1058, 1148, 1172, 1230, 1254
\@tabular@LaTeX 1058, 1254
\@tabular@array 1172, 1272
\@tabular@array@new 1230, 1280
\@tabular@ltx 1148, 1263
\@tabularcr 47
\@tabularcr . . 1073, 1163, 1252, 1260, 1269, 1300
\@tabularcr@LaTeX 47
\@tabularcr@LaTeX . . 1073, 1304
\@tabularcr@ltx 1163, 1305
\@tbpen 47
\@tbpen 1300
\@temp@sw 846, 848, 852
\@tempa 222, 224, 858, 860, 861, 866, 867, 869, 870, 877, 879–882, 906, 907
\@tempb 897, 907
\@tempboxa 66
\@tempc 895, 913
\@tempcnta 35
\@tempcntb 882, 1966, 1969, 1970, 1973, 1975, 1976
\@tempdima 66, 67
\@tempskipa . 929, 931, 932, 937, 963, 964, 1003, 1004
\@temptokena . 1717, 1723, 1748, 1753
\@testdef 380
\@testpach 1650, 1666, 1724, 1755
\@testtrue 1970, 1976
\@textsuperscript 625
\@tfor . . . 1648, 1664, 1722, 1753
\@thefnmark . . 593, 600, 626, 643
\@toodeep 2087
\@totalleftmargin . 2106, 2109, 2113, 2115
\@tpfootnotetext 623
\@trivlist 2099
\@twopowerfourteen 34

\@twopowerfourteen . . . 836, 864
\@twopowertwo 34
\@twopowertwo 837, 881, 882
\@undefined 45
\@undefined 469
\@unexpandable@protect . . 1644, 1660
\@unused 2231, 2237
\@unusedoptionlist . . . 448, 449
\@url 2121, 2122
\@whilesw 1718, 1749
\@width . . 1308, 1309, 1333, 1339, 1391, 1435, 1444, 1466, 1621
\@write@floatline 36
\@write@floatline 903
\@writefile 670
\@x@sf 571, 578
\@xargarraycr 47
\@xargarraycr . 1075, 1165, 1182, 1243, 1300
\@xargarraycr@LaTeX 1075, 1308
\@xargarraycr@array 1182, 1330
\@xargarraycr@ltx . . 1165, 1309
\@xargarraycr@new . . 1243, 1336
\@xarraycr 47
\@xarraycr 1181, 1242, 1300
\@xarraycr@array . . . 1181, 1322
\@xarraycr@new 1242, 1326
\@xbitor 1966
\@xbitor@LaTeX 1966, 1979
\@xbitor@ltx 1972, 1984
\@xfloat 734
\@xfloat@LaTeX 734, 738
\@xfloat@anchored 738, 748
\@xfloat@prep 736, 740
\@xfootnotemark 23
\@xfootnotenext 26
\@xhline 1567, 1573, 1574
\@xhline@unneeded 1576
\@xifempty 13
\@xsect 997, 1025
\@xtabularcr 47
\@xtabularcr . . 1074, 1164, 1300
\@xtabularcr@LaTeX . 1074, 1306
\@xtabularcr@ltx . . . 1164, 1307
\@xvi 835, 845, 846
\@xxxii 841, 842, 844
\@yargarraycr 47
\@yargarraycr . 1076, 1166, 1183, 1244, 1300
\@yargarraycr@LaTeX 1076, 1310
\@yargarraycr@array 1183, 1343
\@yargarraycr@ltx . . 1166, 1311
\@yargarraycr@new . . 1244, 1347
\[. 22, 23
\[. 904
\\ 20, 21, 47
\{ 134, 900, 904
\} 134, 900, 904
\] 22
\] 904
\^ 899, 912, 2153, 2154, 2162
_ 53, 2157, 2174, 2175
\~ 2178
00readme 4
10pt document class option . . . 89
11pt document class option . . . 89
12pt document class option . . . 89

␣ 90, 92
\␣ . . 42, 48, 68, 71, 72, 76, 93, 95, 101, 106, 113, 222, 2127

A

\abovedisplayshortskip . . . 525
\abovedisplayskip 520–525
\active . . . 899, 2154–2159, 2162, 2164, 2167, 2175
\addcontentsline 39
\addcontentsline . 672, 974, 988, 1014, 1022
\addtocontents 660, 673
\AddToHook 345, 409
\aftergroup 53
\aftergroup . . . 1543, 1550, 2064
\AfterLastShipout 16, 18
\appdef 12, 15, 16, 18, 29
\appdef 261, 268, 354, 437, 604, 690, 694, 1040, 1044, 1303, 1413,

1476, 1614, 1638, 1640, 1796, 2227
\appdef@e 286
\appdef@eval 291
\appdef@val 283, 292
aps.rtx 18, 19
argument
class 18
float 36
text 24, 41
your␣tokens␣here 15
argument, optional
number 24
\array 48
\array . . . 1060, 1150, 1174, 1232, 1351
array document class 9, 41, 45, 47, 48, 57, 59
array environment 2, 41, 46
\array@array 1174, 1370
\array@array@new . . . 1232, 1375
\array@default . 806, 1292, 1298, 1541
\array@hook 1050
\array@LaTeX 1060, 1351
\array@ltx 1150, 1359
\array@message 1169, 1246
\array@row@pre . 1416, 1479, 1557
\array@row@pst . 1498, 1504, 1557
\array@row@rst 53
\array@row@rst 1557, 1701, 1710, 1854, 1858, 1862, 1906, 1910, 1914, 1939
\arraycolsep 508, 509, 1371, 1812, 1815
\arrayrulewidth 1566, 1580, 1626
\arraystretch 1389, 1390, 1433, 1434, 1442, 1443, 1464, 1465, 1618, 1620
\AtBeginDocument . 14, 15, 29, 46
\AtBeginDocument 355
\AtEndDocument 15, 16
\AtEndDocument 357, 367
\AtEndOfClass 19
atveryend document class 18, 93
auguide.tex 2
\author 51

B

\baselineskip 12, 22
\baselineskip . 1401, 1424, 1456, 1489
\baselinesklip 12
\begin 15
\begin@float@pagebreak . . . 750, 764, 796, 822
\belowdisplayshortskip . . . 524
\belowdisplayskip 523
\bgroup 26, 53
\body@font 1563, 1598
\botrule 53
\botrule 1562
\box 66

C

\c@mpfootnote 617, 618, 744
\c@secnumdepth . . . 946, 975, 989
\Call@AfterLastShipout 18
\Call@AfterLastShipout 434, 438
\caption 30
\caption 694
\cat@letter 240
\cat@space 2127
\catcode 19
\catcode 478, 486, 899, 904, 2127, 2153–2159, 2162, 2164, 2167, 2175, 2185
\cell@fil 1593, 1636, 1910, 1914
\cell@font 1592, 1599, 1701, 1711, 1906, 1910, 1914, 1939
\centering 20
\changes 138–187
\chardef 239–241, 835, 895, 2127
\check@aux 18
\check@aux 362
⟨*class*⟩ placeholder 18, 19
class, argument 18
\class@documenthook . . . 15, 33
\class@documenthook . 354, 437, 604, 694

\class@enddocumenthook . 15, 29
\class@enddocumenthook 230, 354
\class@err 202
\class@ext@hook 440
\class@extension 440
\class@extensionfile . . 18, 19
\class@extensionfile 440
\class@info 202, 463, 465, 541, 620, 851, 883, 944, 1142, 1144, 1224, 1247, 1250, 1980, 1982, 2218, 2247
\class@name 10
\class@name . 202–204, 223, 232, 233
\class@warn 202, 234
\class@warn@end . . 206, 211, 216, 229
⟨*class customization commands*⟩ placeholder 9
\ClassError 202, 223
classes.dtx 29
classes.dtx document class . . 33
\ClassInfo 204
\classname 62, 70, 122, 124, 126, 141, 165, 166, 178, 179, 181, 182, 186
\classoption 168
\ClassWarningNoLine 203
\clear@document 16–18
\clear@document . . 370, 428, 431
\clearpage 17, 18
\clearpage 424, 432
\closeout 378, 816
\cmd 134
\col@sep 45
\col@sep 1275, 1371
\color 1800
\color@begingroup 645
\color@endgroup 650, 729
colortbl document class . 45, 46, 57, 59, 91
\colortbl@message . . 1169, 1249
\colrule 53
\colrule 1562
\column@fil 1562, 1597
\column@font 1562, 1596
\columnwidth 26
\columnwidth 753
\contentsline 30, 63, 64
\contentsline 674, 691, 692
\contentsline@latex . . 688, 691
\copy 1604
\copyright 48
\count 34, 35
\count . . . 840, 843, 845, 846, 852, 1966, 1973
\count@ . . 1697, 1703, 1719, 1750
\count@i 266
\count@ii 266
\countdef 266, 267
\crcr 1495, 1498, 1501, 1504, 1509, 1585
\cs . 138, 139, 141, 152–154, 156, 157, 159–164, 167, 169, 176, 178, 181–183, 187
\csname 13, 49
\csname 34, 244, 246, 250, 252, 301, 322, 442, 444, 450, 560, 564, 592, 596, 602, 614, 615, 643, 685, 686, 775, 777, 780, 781, 812, 813, 817, 826, 830, 832, 859, 878, 895, 898, 960, 968, 973, 978, 984, 987, 992, 1008, 1013, 1018, 1021, 1240, 1992, 1993, 2032, 2035, 2038, 2040, 2044, 2045, 2053–2055, 2064, 2094, 2217
\CT@cell@color 1803
\CT@color 1798
\CT@column@color 1801
\CT@do@color 1799
\CT@end 46
\CT@row@color 1802
\CT@setup 1797
\CT@start 46
\CurrentOption 455

D

\d@llarbegin . 1253, 1276, 1284, 1372, 1380, 1381
\d@llarend 1277, 1285, 1372, 1381
\deadcycles 372

\DeclareRobustCommand 20
\DeclareRobustCommand 497
\detokenize 413, 414
\dimen 64
\dimen@ . . 863, 864, 866, 867, 869, 870, 1617–1619, 2055, 2056
\dimen@iii 237
\dimendef 237
\discretionary 2125
displaymath environment 22
\displaystyle . 507, 510, 529, 535
\displaywidth 506
\do 1649, 1665, 1723, 1754
\do@check@aux 17, 18
\do@check@aux 362
\do@if@floats 29, 33
\do@if@floats 772
\do@output@cclv 433
doc 4
\DocInput 8
\document 14, 15
\document 343
document class
array 9, 41, 45, 47, 48, 57, 59
atveryend 18, 93
classes.dtx 33
colortbl . . 45, 46, 57, 59, 91
ftnright 9
hyperref 21, 23, 25, 26, 30, 39, 40, 63, 64, 67–69, 89, 91, 93
hyperref␣package 2, 30
hyperref.sty 94
lineno 92, 94
lineno.sty 66, 93
longtable 9, 57
ltxdoc 4, 8
ltxfront 27
ltxgrid . . . 9, 15, 18, 26, 94
ltxutil 9, 18
multicol 89
revtex4 18
revtex4-1 1
textcase 20, 93
document class option
10pt 89
11pt 89
12pt 89
groupauthors 89
hypertext 93
lengthcheck 91
document environment 4, 15
\document@inithook 15
\document@inithook 345, 349, 353, 354, 690, 1040, 1044, 2227
\doi 68
\doi 2118
\doibase 2118
\doublerulesep 1579
\dp 657, 853, 869, 870, 882, 1308, 1309, 1332, 1338, 1390, 1434, 1443, 1465, 1620

E

\edef 57
\edef 271, 276, 281, 413, 414, 571, 1393, 1806, 1812, 1821, 1833, 1880, 2035, 2171–2173, 2181, 2200
\emph 93
\end 36
\end@float@anchored . 751, 752, 758
\end@float@pagebreak 762, 765, 796
\endarray 46, 51
\endarray 1044, 1061, 1151, 1175, 1233, 1494, 1512, 1515, 1518
\endarray@array 1175, 1500
\endarray@array@new 1233, 1503
\endarray@hook 1048, 1049
\endarray@LaTeX 1061, 1494
\endarray@ltx 51
\endarray@ltx . . 1151, 1497, 1504
\endcsname . . . 34, 244, 246, 250, 252, 301, 322, 442, 444, 450, 560, 564, 592, 596, 602, 614, 615, 643, 685, 686, 775, 777, 780, 781, 812, 813, 817, 826, 830, 832, 859, 878, 895, 898, 960, 968, 973, 978, 984, 987, 992, 1008, 1013, 1018, 1021, 1240, 1992, 1993, 2032, 2035,

2038, 2040, 2044, 2045, 2053–2055, 2065, 2094, 2217
\enddocument 17, 18
\enddocument 362
\endgroup 15, 17
\endlinechar 2152
\endminipage 724
\endtabular 1044, 1062, 1152, 1176, 1234, 1240, 1508, 1520
endtabular environment 52
\endtabular* 1520
\endtabular@array . . 1176, 1514
\endtabular@array@new . . . 1234, 1517
\endtabular@hook . . . 1046, 1047
\endtabular@LaTeX . . 1062, 1508
\endtabular@ltx 1152, 1511
\endwrite@float 888, 898
\endwrite@floats 890
environment
array 2, 41, 46
displaymath 22
document 4, 15
endtabular 52
eqnarray 21, 22, 47
equation 22
figure 28
figure@write 28, 33
longtable 29, 36
minipage 23, 28
tabular 2, 41, 46, 47
thebibliography 94
\eqnarray 21
\eqnarray 540, 542, 555
eqnarray environment . 21, 22, 47
\eqnarray@fleqn@fixed 498
\eqnarray@LaTeX 498
\eqncolsep . . . 531, 534, 543, 544
\equation 23
\equation 554
equation environment 22
\errorstopmode 258
\extrarowheight 1440, 1462

F

\f@ur 239, 1743
\false@sw . . . 312, 318, 323, 335, 340, 1079, 1082, 1085, 1088, 1091, 1094, 1097, 1100, 1103, 1106, 1109, 1112, 1115, 1118, 1121, 1124, 1127, 1130, 1133, 1136, 1139, 1186, 1189, 1192, 1195, 1198, 1201, 1204, 1207, 1210, 1213, 1216, 1219, 1222, 2130, 2245
figure environment 28
figure@write environment 28, 33
\figuresname 29
file
.aux 40
.cls 19
.dtx 4
.rtx 19
00readme 4
␣ 90, 92
aps.rtx 18, 19
auguide.tex 2
classes.dtx 29
doc 4
kernel 2, 10, 71
ltfloat.dtx 23
nameref.sty 39
package 2, 9
pdfmark.def 39
README 2
template.aps 2
\file . 29, 81, 84, 90, 91, 100, 106, 108, 111, 112, 115, 116, 122, 124, 126, 128, 131
float, argument 36
\float@end@tag 36
\float@end@tag 903
\floatingpenalty 26, 27
\floats@sw 32, 33
\floats@sw . . . 733, 738, 773, 803
\flushing 490
\fmtversion 303
\font@submax 386, 389
\fontsubfuzz 386
\footins 26

\footins 631, 721
\footmark 24
\footnote 23, 24, 28
\footnote 557, 664
\footnote@latex 605
\footnotemark 23, 24, 26
\footnotemark 557
\footnotesep 647, 656
\footnotesize 654
\footnotetest 557
\footnotetext 23, 24
\footnotetext 581
\frontmatter@footnotetext . 27
\frontmatter@makefntext . . . 27
\frstrut 1562, 1563, 1600
ftnright document class 9
\ftype@ 34
\fullinterlineskip . . . 265, 711
\futurelet 1566, 1573

G

\gappdef 230, 268
\GetFileInfo 39
\glet@environment 217, 242
\glossary 663
groupauthors document class option 89

H

\H@@footnotetext 23
\H@@footnotetext . 607, 608, 745
\H@@mpfootnotetext 609, 610, 745
\H@refstepcounter 38
\H@refstepcounter . . . 950, 1039
\H@svsec 947, 951, 969, 985, 1002, 1009, 1019
\halign 51, 57
\halignt@ 238, 528
\hangindent 67
\hb@xt@ 511, 536, 769, 2062, 2074
\hbox 22, 67
\hline 54
\hline 1072, 1162, 1565
\hline@LaTeX 1072, 1565
\hline@ltx 1162, 1569
\hline@rule 54
\hline@rule . . 1564, 1572, 1587, 1602
⟨*hooks and bookkeeping*⟩ placeholder 17
\href 68, 69
\href 2118, 2148
\href@ 2148–2150
\href@Hy 2148, 2217
\href@Hy@ltx 2149, 2224
\hrule 12
\hsize 14
\ht 853, 863, 866, 867, 881, 1389, 1433, 1439, 1461, 1617, 1618
\Hy@ActiveCarriageReturn 2189, 2222
\Hy@ActiveCarriageReturn@ltx 2186, 2222
\Hy@footnote@currentHref . 574, 615
\Hy@href 2149, 2150
\Hy@raisedlink 1032
\Hy@RemovePercentCr 2202, 2211, 2223
\Hy@RemovePercentCr@ltx . 2206, 2223
\Hy@ReturnAfterElseFi . . . 2190
\Hy@ReturnAfterFi . . 2194, 2210
\Hy@safe@activestrue . . . 2182
\Hy@tempa 2188, 2189, 2200, 2204
\hyper@@normalise . 2191, 2195, 2220
\hyper@@normalise@ltx . . . 2199, 2220
\hyper@anchorend 40
\hyper@anchorend 1034
\hyper@anchorstart 40
\hyper@anchorstart 1033
\hyper@hash 2168, 2169
\hyper@last 40
\hyper@linkend 40
\hyper@linkend 576, 1036
\hyper@linkstart 40
\hyper@linkstart 574, 1035
\hyper@makecurrent 1031
\hyper@n@rmalise . . . 2183, 2221

\hyper@n@rmalise@ltx 2187, 2221
\hyper@normalise . . . 2148–2150, 2219
\hyper@normalise@ltx 2160, 2219
\hyper@tilde 2177–2179
\hyperanchor 40
hyperref document class . 21, 23, 25, 26, 30, 39, 40, 63, 64, 67–69, 89, 91, 93
hyperref␣package document class 2, 30
hyperref.sty document class . 94
hypertext document class option 93
\hypertext@enable@ltx 68
\hypertext@enable@ltx . . . 2128

I

\ialign 49
\ialign 1394, 1447
\if 49
\if 309, 321, 1386, 1452
\if@filesw . 377, 394, 1991, 2020
\if@firstamp 1806, 1809
\if@nobreak 934
\if@noskipsec 927
\ifeof 325
\iffalse 55, 1314, 1319
\IfFileExists 441
\IfFormatAtLeastTF . . . 302, 305
\ifhbox 326
\ifinner 328
\ifmmode 329
\ifodd 331, 1970
\ifvbox 332
\ifvoid 334
\ignorespaces 647, 1526, 1539, 1696, 1702, 1883, 1885, 1887, 1906, 1910, 1914, 1931, 1939, 2061, 2070, 2082, 2102
\immediate . . 378, 778, 815, 913, 1993, 2043, 2231, 2237
\incompatible@package 220
\index 662
\init@hyperref 1029
\insert 27
\insert 631, 721
\insert@column . 1179, 1239, 1694
\insert@column@array 1179, 1694
\insert@column@array@new 1239, 1699
\interdisplaylinepenalty . . 47
\interfootnotelinepenalty 655
\interlinepenalty 655, 966, 1006
\interlineskip 27
\intertabularlinepenalty . . 47
\intertabularlinepenalty 1300, 1301, 1303, 1305, 1320
\intextsep 764, 765
\item 104, 114, 117, 121, 123, 125, 130, 132
\itemindent 2093
\itshape 617, 626

J

\j@nk 341

K

kernel 2, 10, 71

L

\l@ 64
\l@@sections 64
\l@@sections 2049
\l@section 2003
\label 661, 672
\lastbox 551, 714, 820
\LaTeX . 42, 71, 72, 82, 92, 93, 95, 106, 110, 113, 115, 187
\LaTeXe 76, 101, 197
\leftmargin . . . 2106, 2108, 2113
\leftmargini 2003
\leftmarginii 2004
\leftmarginiii 2005
\leftmarginiv 2006
\leftmarginv 2007
\leftmarginvi 2008
\leftskip 67
\leftskip . 492, 2054, 2056, 2115
lengthcheck document class option 91
\lengthcheck@sw 798

\let 45, 65
\let@environment 11
\let@environment . 242, 788, 789, 791, 810, 811
\limits 2208
lineno document class . . . 92, 94
lineno.sty document class 66, 93
\lineskip 1401, 1424, 1455, 1489
\linewidth 528, 2107–2109
\list 67
\list 2085
\listparindent . 2092, 2105, 2112
\LoadClass 19
\long 71
longtable document class . 9, 57
longtable environment . . 29, 36
\loop 14
\loopuntil 337
\loopwhile 337
\lrstrut 1563, 1564, 1601
\LT@array 9
ltfloat.dtx 23
\ltx@contentsline 672
\ltx@def@footproc 24, 25
\ltx@def@footproc 559, 567, 582, 590
\ltx@footmark 24
\ltx@footmark 557
\ltx@footnote@pop 741, 747
\ltx@foottext 24
\ltx@foottext 557
\ltx@make@current@footnote 557
\ltx@mathindent 22
\ltx@mathindent 518, 548
\ltx@stp@footproc 24, 25
\ltx@stp@footproc 563, 568, 583, 598
\ltx@thempfootnote . . . 618, 621
\ltx@xfootmark 566, 567
\ltx@xfootnote 24
\ltx@xfootnote 557
\ltx@xfoottext 581, 582
\ltx@yfootmark 566, 568
\ltx@yfootnote 557, 562
\ltx@yfoottext 581, 583
ltxdoc document class 4, 8
ltxfront document class 27
ltxgrid document class 9, 15, 18, 26, 94
\ltxu@dotsep 2002, 2077
ltxutil document class . . . 9, 18

M

\make@footnotetext 26, 27
\make@footnotetext 623
\makeatletter . 32, 382, 454, 1989
\makeatother 35
\makelabel 2096
\maketitle 26
\maketitle 60
\mark 29
\mathchardef 836, 837
\mathindent 22
\meaning 263
\MessageBreak 388
minipage environment . . . 23, 28
\minipagefootnote@drop 26
\minipagefootnote@drop 651, 697
\minipagefootnote@foot . . . 697
\minipagefootnote@here 31
\minipagefootnote@here . . . 695, 697, 759
\minipagefootnote@init 697, 756
\minipagefootnote@pick 635, 697
\mpfootnote 620
multicol document class 89
\multicolumn . . 1057, 1147, 1521
\multicolumn@LaTeX . 1057, 1521
\multicolumn@ltx . . . 1147, 1528
\multiply 842
\multispan 1522, 1529

N

nameref.sty 39
\NC@list 1718, 1749
\newbox 1613
\newcount 1300, 1302
\newenvironment 36
\newlabel 30
\newlabel 680
\newlinechar 912
\newwrite 774, 1992

\noalign . 1307, 1309–1311, 1328, 1341, 1345, 1349, 1564, 1566, 1570, 1574, 1586
\noexpand . 263, 264, 1394, 1447, 2202
\normalfont 626
number, optional argument . . . 24
\numberline 66, 67
\numberline@@sections . . . 2059, 2073

O

\obeylines 909
\obsolete@command 205
\oneapage 796
\openout 779, 1993
\OptionNotUsed 19
\other 241

P

\p@equation 500, 516
package 2, 9
\package@name 195, 196
\PackageInfo 196
\par 66, 71
\parindent 22, 67
\parindent 767, 2056, 2074, 2105, 2112
\parsep 2100
\parshape 67
\parshape 2109
\parskip 22
\parskip 522, 2100
\parskipfillskip 22
\partopsep 521
\pbox@hook 1637, 1939
\pdfendlink 2141
pdfmark.def 39
\pdfoutput 2130
\pdfstartlink 2137
\pdfstartlink@attr 68
\pdfstartlink@attr 2128
\pdfstringdefDisableCommands 28
\phantomsection 39
\phantomsection . 947, 952, 1002, 1030
placeholder
⟨*class*⟩ 18, 19
⟨*class customization commands*⟩ 9
⟨*hooks and bookkeeping*⟩ . . 17
⟨*read main .aux and final checks*⟩ 17
⟨*your document here*⟩ 8
\prep@math 22
\prep@math 550
\prep@math@patch 22, 23
\prep@math@patch 550
\prepdef 12
\prepdef 268, 347, 554, 555, 695, 812, 813, 956, 958, 1046, 1048, 1051, 1409, 1472
\prepnext@tok 1721, 1752
\prevdepth 12
\prevdepth 265
\print@float 33
\print@float 795
\printfigures 29
\ProcessOptions 19
\protected 411
\protected@edef 25
\protected@edef . . 615, 642, 954
\protected@write 670, 679
\protected@xdef 600
\providecommand 438, 1030–1037, 1050, 2118, 2121, 2123, 2124, 2126, 2127
\ProvidesFile 6
\ProvidesPackage 19

R

\raggedleft 20
\raggedright 20
⟨*read main .aux and final checks*⟩ placeholder 17
README 2
\realfootnote 26
\RecordChanges 37

\refstepcounter 38
\relax 38, 41
\repeat 14
\replace@command 210
\replace@environment 215
\RequirePackage 25, 27, 34
\reserved@a . . 1567, 1573, 1577, 1578, 1649, 1664
\reset@font 654
revtex4 document class 18
revtex4-1 document class 1
\robustify@contents 660
\romannumeral 2094
\rule 647
\rvtx@enddocument@patch . . 408
\rvtx@enddocument@patch@end 417, 424
\rvtx@enddocument@patch@more 419, 422
\rvtx@ifformat@geq 13
\rvtx@ifformat@geq . . 302, 343, 362, 408, 2244

S

\samepage 47
\samepage 1303
\save@decl 1726, 1763
\say 11
\say 263, 1577
\saythe 11
\saythe 263
\sc 30, 92
\sec@upcase 41
\sec@upcase 1043
\section 88, 823
\Sectionformat 39
\set@arstrutbox 54
\set@arstrutbox 1407, 1431, 1602
\set@display@protect 2230, 2236
\set@eqnarray@skips 549
\set@footnotefont 28
\set@footnotefont 639, 653
\set@footnotewidth 26, 28
\set@footnotewidth 623
\set@footnotewidth@one . . . 659
\set@listindent 2101, 2117
\set@listindent@ 2111
\set@listindent@parshape 2104, 2117
\set@tocdim@pagenum . . . 66, 67
\set@tocdim@pagenum 2052, 2069
\set@typeset@protect . . . 1400, 1423, 1488, 1524, 1533
\shipout 11
\show@box@size@sw 850
\showboxbreadth 258
\showboxdepth 258
\showoutput 261
\skip@ 2058, 2061
\spacefactor 571, 578
\special 2132, 2133
\splitmaxdepth 657
\splittopskip 656
\stepcounter 499, 516, 599
\stop 224
\StopEventually 5
\string 206, 211, 222, 620, 665–667, 670, 680, 944, 1980, 1982, 2044, 2171–2173, 2181, 2218, 2247
\strutbox 648, 657, 1389, 1390, 1433, 1434, 1439, 1443, 1461, 1465, 1618, 1620
\subsection 98, 137
\substyle@ext 19
\switch@array 41, 47
\switch@array 1045, 1054
\switch@array@info . 1169, 1226
\switch@hyperref@href . . . 2216, 2227
\switch@tabular 41, 47
\switch@tabular 1045, 1054

T

\t@ 236, 238
\tab@rule 54
\tab@rule 1562, 1563, 1584
\tabcolsep 1275, 1633–1635, 1821
\tableft@skip . . 1425, 1490, 1630
\tableftsep . . . 1633, 1707, 1818
\tableofcontents 86

\tabmid@skip . . 1410, 1473, 1631
\tabmidsep 1634, 1708, 1824
\tabright@skip . 1414, 1477, 1632
\tabrightsep . . 1635, 1709, 1827
\tabskip 22, 49
\tabskip 504, 507, 510, 512, 518, 529, 535, 537, 1395, 1410, 1414, 1425, 1449, 1473, 1477, 1490
\tabular 41
tabular environment 2, 41, 46, 47
\tabular@hook 1051, 1052
\tabularnewline 1397, 1420, 1458, 1485
\tally@float 34
\tally@float 835
template.aps 2
\TeX 68, 76, 115, 186
text, argument 24, 41
\textasciitilde 2179
textcase document class . 20, 93
\texttt . . 53, 111, 112, 119, 128, 131, 134, 138
\textunderscore 2173, 2174, 2176
\textwidth 865
\thanks 43, 47, 53
\the@toks 1695, 1697, 1700, 1703, 1720, 1739, 1751, 1794
thebibliography environment 94
\theequation 500, 516
\thefootnote 23
\thempfn 23
\thempfn 593, 600, 743
\thempfootnote 25
\thempfootnote . . . 619, 621, 743
\thempfootnote@latex . 617, 619
\thepage 674, 680
\thr@@ 11
\thr@@ . 237, 511, 536, 1735, 1786
\title 41
\toc@@font 64
\toc@@font 2001, 2010
\toc@letdimen . . 2016, 2017, 2037
\toc@post 65
\toc@post 1996, 2000
\toc@post@auto 65
\toc@post@auto 2009
\toc@pre 65
\toc@pre 1988, 1999
\toc@pre@auto 65
\toc@pre@auto 2009
\toc@setindent . 2012–2015, 2031
\toc@writedimen 2022–2027, 2042
\tocdim@appendix 2007
\tocdim@min . . . 2032, 2038, 2084
\tocdim@pagenum 2008, 2071
\tocdim@paragraph 2006
\tocdim@section 66
\tocdim@section 2003
\tocdim@subsection 2004
\tocdim@subsubsection . . . 2005
\tocleft@pagenum 2057
\tocleft@section 66
\tocmax@section 66
\toks 1739, 1794
\toks@ . . . 269, 271, 274, 276, 279, 281, 896, 913, 915
\toks@ii 12
\toks@ii . 270, 271, 275, 276, 280, 281, 296
\toksdef 296
\toprule 53
\toprule 1562
\topsep 520
\total@float 34
\total@float 799, 835
\traceoutput 11
\traceoutput 260
\tracingall 11
\tracingcommands 255
\tracinglostchars 256
\tracingmacros 257
\tracingonline 255
\tracingoutput 11
\tracingoutput 256
\tracingpages 256
\tracingparagraphs 257
\tracingplain 11
\tracingplain 254
\tracingrestores 257
\tracingstats 255

\trigger@float@par .. 809, 812, 813, 819
\triggerpar 795, 809
\true@sw 310, 317, 323, 335, 340, 350, 1077, 1184, 2130, 2242, 2245
\tw@ 11
\typeout 71
\typeout 2228
\typeout@ltx 2234, 2241
\typeout@org 2228, 2240

U

\unhbox 2062, 2079
\unhcopy 1604
\unrestored@protected@xdef 593
\unskip 726, 1331, 1337, 1563, 1564, 1696, 1702, 1883, 1885, 1887, 1906, 1910, 1914, 2061
\unvbox 702, 716, 721, 731
\unvcopy 713, 862
\uppercase 41
\url 68
\url 26, 66, 73, 2118
\URL@prefix 2118

V

\vadjust 648
\vbox .. 701, 862, 1386, 1452, 1542
\vcenter 53
\vcenter . 1386, 1452, 1543, 1547, 1550
\voidb@x 698
\vrule 1308, 1309, 1333, 1339, 1389, 1433, 1441, 1464, 1619
\vsize 14
\vsplit 27
\vtop 713, 1386, 1452, 1541

W

\write 913, 2043, 2231, 2237
\write@@float 887, 891
\write@float 29, 33, 36
\write@float 785, 887
\write@floatline 36
\write@floatline 901, 903
\write@floats 786, 887

X

\x 413, 416
\xdef 458, 474, 1447, 1470

Y

\y 414, 416
your␣tokens␣here, argument . 15
⟨*your document here*⟩ placeholder 8

Z

\z@skip .. 492–494, 507, 512, 529, 537, 648, 760, 1395, 1401, 1424, 1630–1632, 2062

Change History

4.0a

General: 4.0d had twoside option setting twoside switch to false . 2

comma not space between email and homepage 2

Initial version 2

Move after process options, so `\clearpage` not in scope of twocolumn 2

Move after process options, so the following test works 2

multiple preprint commands . . . 2

print homepage 2

protect against hyperref revtex kludges which are not needed now 2

Rearrange the ordering so numerical ones come first. AO: David, what does this mean? . . 2

single space footnotes 2

use font-dependent spacing 2

4.0b

`\@mpmakefntext`: AO: Removed superfluous `\def`s, changed to using `\floats@sw` as the flag. Also stopped using DPC's `\if@twocolumn` flag: using `\floats@sw` instead. Also added `\par\vskip\z@skip` after the `\minipagefootnotes` so that the float box would have zero depth like the kernel one. 32

`\caption`: Support the hack with `\prepdef`, and delay until `\AtBeginDocument` time, since `hyperref` clobbers `\caption`. . 30

`\print@float`: only execute if there really were floats of the given type 33

`\write@@float`: AO: Fixed spurious `CR` and (return) characters in output file. Also, if the document did not have the `\endfigure` on a line of its own, the macro wouldn't work. Fixed. 36

General: Added localization of `\figuresname` 2

Added localization of `\tablesname` 2

AO: all code for `10pt` is in this module. 2

AO: all code for `11pt` is in this module. 2

AO: all code for `12pt` is in this module. 2

AO: made aps.rtx part of revtex4.dtx 2

AO: remove duplicates 2

call `\print@floats` 2

Defer assignment until `\AtBeginDocument` time. 2

Defer decision until `\AtBeginDocument` time 2

Define three separate environments, defer assignment to `\AtBeginDocument` time. . . . 2

First modifications by Arthur Ogawa (mailto:arthur_ogawa at sbcglobal dot net) 2

Frank Mittelbach, has stated in `multicol`: "The kernel command `\@footnotetext` should not be modified." Thus, I have removed David Carlisle's redefinition of that command. Note, however, that later versions of `multicol` do not require this workaround. Belt and suspenders. 2

Move this "complex" option to the front, where it can be overridden by "simple" options. 2

New option 2

One-line caption sets flush left. . 2

only execute if appropriate 2

Processing delayed to `\AtBeginDocument` time 2

Removed invocation of nonexistent class option `groupauthors` and all other class options that should only be invoked by the document.

(Otherwise precedence of class options does not work.) 2
Restore all media size class option of ␣classes.dtx 2
Stack \preprint args flush right at right margin. 2
4.0c
\@mpmakefntext: (AO, 110) Install hooks for endfloats processing 32
\@ssect: (AO, 116) Hyperref compatibility 39
\endarray: (AO, 130) Interference from array package 41
\print@float: *-form mandates pagebreak at each float; only print section head if there is something there. 33
General: (AO, 115) If three or more preprints specified, set on single line, with commas. 2
(AO, 129) section* within appendix was producing appendixname 2
*-form mandates pagebreak . . . 2
also spelled "acknowledgements". 2
Do not put by REVTeX in every page foot 2
grid changes via ltxgrid procedures 2
grid changes with ltxgrid 2
Insert procedure \checkindate . 2
Lose compatability mode. 2
New ltxgrid-based code, other bug fixes 2
New option "checkin" 2
Prevent an inner footnote from performing twice 2
4.0d
\@mpmakefntext: (AO, 127) Floats placed [h] to allow page breaks 32
(AO, 224) Hyperref compatibility. 32
\print@float: Allow things to break over pages by setting array@default. 33
General: (AO, 174) kernel fix . . . 23
Also alter how lists get indented. 2
But alternative spelling is deprecated. 2
eprint takes an optional argument, syntactical only in this case. 2
make longtable trigger the head, too 2
More features and bug fixes: compatability with longtable and array packages. Now certainly incompatible with multicol. 2
New option 2
4.0e
\@mpmakefntext: (AO, 221) Remove samepage command from @xfloat@prep: If the float can break over pages, we want better control. 32
General: adornments above and below. 2
Bug fixes and minor new features: title block affiliations can have ancillary data, just like authors; clearpage processing revamped, with floats staying in order; widetext ornaments. 2
New option showkeys 2
4.0f
\@ssect: (AO, 404) Hyperref compatibility 39
General: Last bug fixes before release. 2
4.0rc1
General: grid changes with push and pop 2
Running headers always as if two-sided 2
4.0rc4
General: hyperref is no longer loaded via class option: use a usepackage statement instead . 2
4.1a
\@mpmakefntext: \@xfloat@prep calls \ltx@footnote@pop to restore the original \ltx@footmark and \ltx@foottext procedures, in case footnote processing has switched. 32

\@p@pfilename: Class extension mechanism \@pushfilename@ltx and \@p@pfilename@ltx. 19
\class@enddocumenthook: \class@documenthook is the last \AtBeginDocument token now 15
\document: Get rid of \set@typesize@hook \set@pica@hook and the \normalsize directive 15
\eqnarray@fleqn@fixed: (AO, 475) I had not properly reproduced the LaTeX macro \eqnarray. 21
\footnote: (AO) Remove code that avoided changes to \@xfootnotemark 26
\ltx@make@current@footnote: (AO, 438) Complete rewrite of footnote macros. 23
\numberline@@sections: (AO, 461) Change the csname from \@dotsep to \ltxu@dotsep. The former is understood in mu. (What we wanted was a dimension.) 67
\robustify@contents: (AO) Make \addtocontents a \long \def; gobble up \footnote 29
\toc@@font: (AO, 461) Change the csname from \@dotsep to \ltxu@dotsep. The former is understood in mu. (What we wanted was a dimension.) . . . 64
General: (AO, 451) "Cannot have more than 256 cites in a document" 2
(AO, 457) Endnotes to be sorted in with numerical citations. . . . 2
(AO, 459) do not assume \class@name is defined 10
(AO, 460) "Proper style is "FIG. 1. ..." (no colon)" 2
(AO, 461) Change the csname revtex uses from @dotsep to ltxu@dotsep. The former is understood in mu. (What we wanted was a dimension.) 2
(AO, 478) \ds@letterpaper, so that "letterpaper really is the default" 2
(AO, 488) Change processing of options to allow an unused option to specify society and journal 2
Class extension mechanism \class@extension, \class@extensionfile, and \class@ext@hook. 18
For natbib versions before 8.21, \NAT@sort was consulted only as natbib was being read in. Now it is fully dynamic. 2
4.1b
\@mkpream: (AO, 505) try to accommodate colortbl. 57
\@mkpream@relax: (AO, 505) try to accommodate colortbl. . . 59
\@mpmakefntext: No need to protect against undefined \float@sw 32
\@tabarray: (AO, 505) try to accommodate colortbl. 46
\@tabular: (AO, 505) try to accommodate colortbl. 45
\array: (AO, 505) try to accommodate colortbl. 49
\do@if@floats: No need to protect against undefined \float@sw . 33
\doibase: (AO, 487) Support for video figures and the \setfloatlink command . . . 68
\endarray: Patch the array package even later: after all package patches go in. 41
\floats@sw: Default assignment of \float@sw now, not at \AtBeginDocument time. 32
\init@hyperref: Acquire hyperref savoire 40
\ltx@contentsline: Refine toc processing: provide default. . . 30
\print@float: If class option lengthcheck is in effect, log the height of this float class. . 33
\switch@array: (AO, 505) Try to accommodate colortbl. 44

\total@float: Tally and log the height of a float class 34
General: (AO) Implement bibnotes through \frontmatter@footnote@produce instead of \bibnotes@sw 2
(AO) No longer need to test \chapter as of natbib version 8.2 2
(AO) No longer use \secnumarabic@sw, instead use \setup@secnums 2
(AO) Provide more diagnostics when \@society is assigned. . . 2
(AO) Structure the Abstract using the bibliography environment 2
(AO) coordinate \if@twoside with \twoside@sw 2
(AO) make settings at class time instead of deferring them to later. 2
(AO) provide option longbibliography 2
(AO, 455) Be nice to a list within the abstract (assign \@totalleftmargin). 2
Add \@hangfroms@section 2
Add option reprint, opposite of preprint, and preferred alternative to twocolumn 2
As with author formatting, rag the right more, and assign \@totalleftmargin. Also neutralize \def@after@address. 2
Break out \@caption@fignum@sep 2
Class option galley sets \preprintsty@sw to false 2
Code relating to new syntax for frontmatter has been placed in ␣ ltxfront.dtx 2
Package textcase is now simply a required package 2
Procedures \@parse@class@options@society and \@parse@class@options@journal and friends 2
Rag the right even more: .8\hsize. Also, assign \@totalleftmargin. 2
Read in all required packages together 2
Remove options newabstract and oldabstract 2
Section numbering via procedures \secnums@rtx and \secnums@arabic. 2
The rmp journal substyle selects groupedaddress by default. . . . 2
The csname substyle@ext is now defined without a dot (.), to be compatible with LaTeXusage (see @clsextension and @pkgextension). 2
Use \setup@hook to initialize all. 2
4.1c
General: Document class option longbibliography via \substyle@post 2
4.1d
\eqnarray@fleqn@fixed: (AO, 511) Compatability with lineno.sty's erroneous way of detecting fleqn.clo 22
General: Definition of \@fnsymbol follows fixltx2e.sty 2
4.1e
General: (AO, 455) be nice to a list within the abstract 2
4.1f
\set@footnotewidth: (AO, 515) Hook for setting the font of a footnote 28
\total@float: (AO, 518) Tally register overflow when locument is long 34
General: (AO, 513) Add class option linenumbers: number the lines a la lineno 2
(AO, 516) Merged references are separated with a semicolon . . . 2
(AO, 520) Automatically produce \bibliography command when needed 2
(AO, 521) Lonely bibliography head 2

(AO, 522) Warn if software is expired 2
(AO, 523) Add class option nomerge, to turn off new natbib 8.3 syntax 2
(AO, 524) Makes no sense if citations are superscript numbers and so are footnotes . 2
(AO, 530) \@fnsymbol: Failed to import fixltx2e.sty technology. Return to LaTeX core. 2
4.1g
\doibase: (AO, 532) Both arguments of \href get sanitized 68
General: (AO, 525) Remove phantom paragraph above display math that is given in vertical mode 2, 22
(AO, 538) \MakeTextUppercase inappropriately expands the double backslash 2
(AO, 539) Use of double-backslash in argument of \section gives error. The textcase package is involved. 20
4.1h
General: (AO) Remove expiry code in the release software 2
4.1i
General: (AO, 541) Defer assignment of \cite until after natbib loads 2
4.1j
\doibase: (AO, 545) Provide definition for \doi that does hypertext 68
\hypertext@enable@ltx: (AO, 545) hypertext capabilities off by default; enable with hypertext 68
General: (AO, 545) hypertext capabilities off by default; enable with hypertext 2
(AO, 549) Repairing natbib's \BibitemShut and \bibAnnote 2
(AO, 552) Repair spacing in \onlinecite 2
4.1k
General: (AO, 554) give the \newlabel command syntax appropriate to the hyperref package 2
4.1n
\clear@document: (AO, 569) Use of hyperref interferes with column balancing of last page 18
(AO, 569) execute atveryend's \Call@AfterLastShipout at the proper time 18
\do@check@aux: (AO) Incorporate change to ltmiscen.dtx v1.1i 2000/05/19 16
(AO, 569) Use of hyperref interferes with column balancing of last page 15
\l@@sections: (AO, 574) protect against lineno.sty, which forces a visit to the output routine, which appears to destroy the value of \@tempdima 66
\set@footnotewidth: (AO, 571) Interface \set@footnotewidth for determining the set width of footnotes 28
(AO, 571) allow split after last line of footnote 28
(AO, 572) title block footnotes numbered independently from body footnotes 27
General: (AO) fine-tune spacing above and below widetext 2
(AO, 565) restore 4.0 behavior: invoking class option preprint implies class option preprintnumbers 2
(AO, 566) restore 4.0 behavior: flush column bottoms 2
(AO, 569) Use of hyperref interferes with column balancing of last page 2
(AO, 569) execute the after-last-shipout procedures from within the safety of the output routine 2
(AO, 571) Interface \set@footnotewidth for determining the set width of footnotes 2

(AO, 571) class file must set `\splittopskip`; fine tune `\skip\footins`; `\footnoterule` defined in terms of `\skip\footins` 2
(AO, 572) Independent footnote counter for title block. Abstract footnote counter shared with body. 2
(AO, 572) `\@makefntext` and `\frontmatter@makefntext` must be defined harmoniously . 2
(AO, 573) arrange to load `lineno` after any other packages. 2
(AO, 575) the default for journal prstper is longbibliography . . . 2
(AO, 576) In .bst files, remove support for the annote field . . . 2
4.1o
General: (AO, 549) Remove patch to natbib, which is now at version 8.31a 2
(AO, 575) Automatically incorporate the (BibTEX-generated) .bbl into an explicit `thebibliography` . . 2
(AO, 578) accommodate the possible space character preceding `\BibitemShut`. 2
(AO, 579) Endnote shall comprise their own BibTEX entry type: @FOOTNOTE. . . . 2
(AO, 580) Control .bst at run time. 2
(AO, 580) Provide a document class option to turn off production of eprint field in bibliography. 2
(AO, 581) Handle case: merged references, with first ending in a stop character. 2
4.1p
`\href`: (AO, 582) A patch of `hyperref.sty` to provide backward compatibility to TEXLive 2007's version 6.75r . 69
General: (AO, 583) Provide interface to `ltxgrid` `\onecolumn@grid@setup` and `\twocolumn@grid@setup` 2
(AO, 584) Per MD, remove trailing space character from each journal abbreviation: it had caused an extraneous space in the .bbl 2
4.1q
General: (AO, 586) When .bbl is pasted into the document, prevent automatic bibliography inclusion. 2
(AO, 588) Only write REVTEX-specific BibTeX .bib data if the .bst style is set by REVTeX. 2
4.1r
General: (AO, 595) Provide `\lovname` along with other List of Videos definitions. 2
4.1s
General: (PHO) Adapt patches to `\document` and `\enddocument` to the 2020-10-01 LATEX release. 2
4.2d
`\do@check@aux`: (PHO) Only redefine `\enddocument` in older versions. 15
(PHO) Patch `\enddocument` at runtime in newer versions. . . . 17
`\document`: (PHO) Use LATEX's hook management system, if possible (from 4.2). 15
`\rvtx@ifformat@geq`: (PHO) Add `\rvtx@ifformat@geq` (from 4.2). 13